2.04 Trigonometry: Trigonometric Identities in Radians

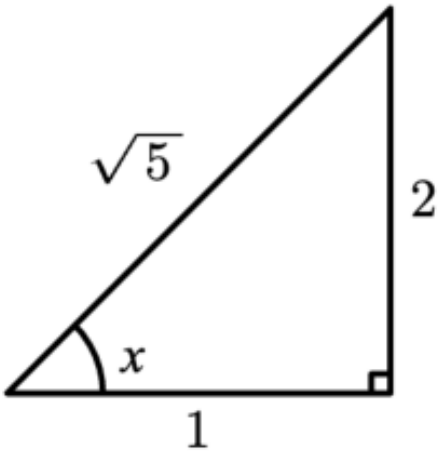
Worked Solutions

1 a)

$\sin^2 x = \frac{4}{5} \Rightarrow \sin x = \pm\frac{2}{\sqrt{5}}$.

x is **ACUTE** so $\sin x = \frac{2}{\sqrt{5}}$ (the result is positive as $\sin x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

Draw a triangle with this information:



$\cos x = \frac{1}{\sqrt{5}}$ (the result is positive as $\cos x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

1 b)

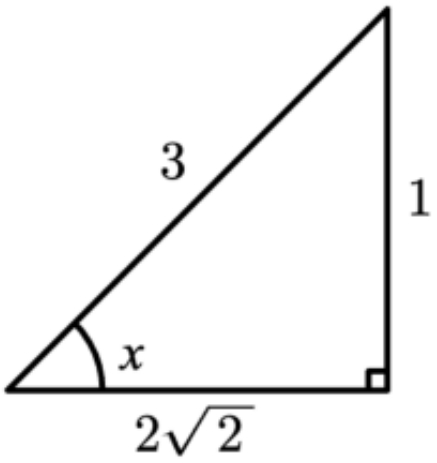
$\tan x = \frac{2}{1} = 2$ (the result is positive as $\tan x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

2 a)

$\cos^2 x = \frac{8}{9} \Rightarrow \cos x = \pm\frac{2\sqrt{2}}{3}$.

x is **ACUTE** so $\cos x = \frac{2\sqrt{2}}{3}$ (the result is positive as $\cos x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

Draw a triangle with this information:



$\sin x = \frac{1}{3}$ (the result is positive as $\sin x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

2 b)

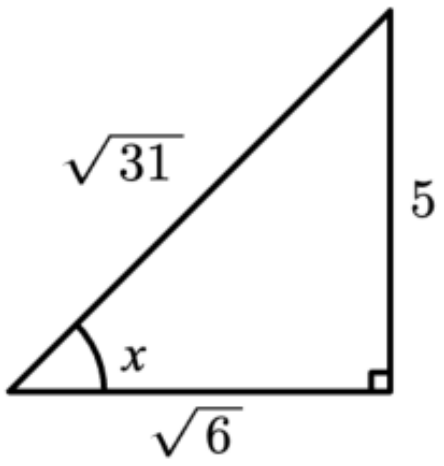
$\tan x = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$ (the result is positive as $\tan x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

3 a)

$\tan^2 x = \frac{25}{6} \Rightarrow \tan x = \pm\frac{5}{\sqrt{6}}$.

x is **ACUTE** so $\tan x = \frac{5}{\sqrt{6}}$ (the result is positive as $\tan x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

Draw a triangle with this information:



$\sin x = \frac{5}{\sqrt{31}}$ (the result is positive as $\sin x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

3 b)

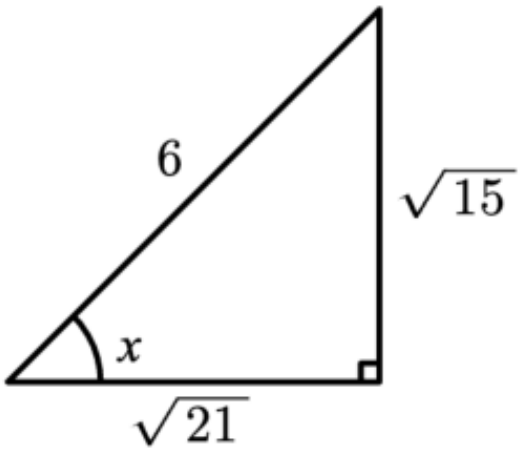
$\cos x = \frac{\sqrt{6}}{\sqrt{31}} = \frac{31\sqrt{6}}{31}$ (the result is positive as $\cos x$ is above the x-axis between 0 and $\frac{\pi}{2}$).

4 a)

$\sin^2 x = \frac{5}{12} \Rightarrow \sin x = \pm\frac{\sqrt{5}}{\sqrt{12}} = \pm\frac{\sqrt{5}}{2\sqrt{3}} = \pm\frac{\sqrt{15}}{6}$.

x is **OBTUSE** so $\sin x = \frac{\sqrt{15}}{6}$ (the result is positive as $\sin x$ is above the x-axis between $\frac{\pi}{2}$ and π).

Ignore $\frac{\pi}{2} < x < \pi$ for the moment and draw a right-angled triangle with this information:



$\cos x = -\frac{\sqrt{21}}{6}$ (the result is negative as $\cos x$ is below the x-axis between $\frac{\pi}{2}$ and π).

4 b)

$$\tan x = -\frac{\sqrt{15}}{\sqrt{21}} = -\frac{\sqrt{15}\times\sqrt{21}}{21} = -\frac{\sqrt{3}\times\sqrt{5}\times\sqrt{3}\times\sqrt{7}}{21}$$
$$= -\frac{3\sqrt{35}}{21} = -\frac{\sqrt{35}}{7}.$$

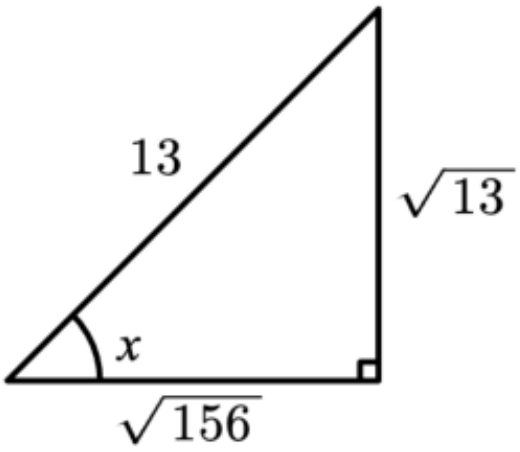
(the result is negative as $\tan x$ is below the x-axis between $\frac{\pi}{2}$ and π).

5 a)

$\cos^2 x = \frac{12}{13} \Rightarrow \cos x = \pm\frac{\sqrt{12}}{\sqrt{13}} = \pm\frac{\sqrt{12}\times\sqrt{13}}{13} = \pm\frac{\sqrt{156}}{13}$.

x is **OBTUSE** so $\cos x = -\frac{\sqrt{156}}{13}$ (the result is negative as $\cos x$ is below the x-axis between $\frac{\pi}{2}$ and π).

Ignore $\frac{\pi}{2} < x < \pi$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = \frac{\sqrt{13}}{13}$ (the result is positive as $\sin x$ is above the x-axis between $\frac{\pi}{2}$ and π).

5 b)

$$\tan x = -\frac{\sqrt{13}}{\sqrt{156}} = -\frac{\sqrt{13}\times\sqrt{156}}{156} = -\frac{\sqrt{13}\times\sqrt{13}\times\sqrt{12}}{156}$$
$$= -\frac{13\sqrt{12}}{156} = -\frac{\sqrt{12}}{12} = -\frac{2\sqrt{3}}{12} = -\frac{\sqrt{3}}{6}.$$

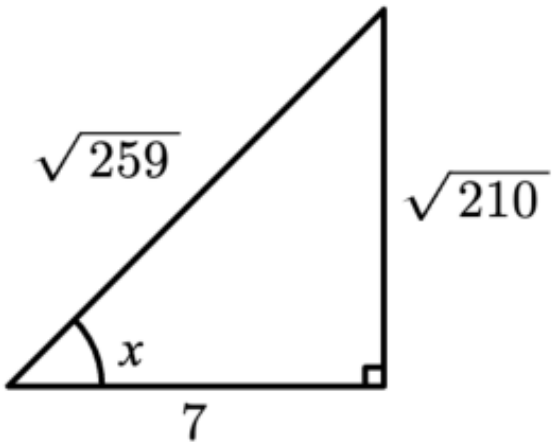
(the result is negative as $\tan x$ is below the x-axis between $\frac{\pi}{2}$ and π).

6 a)

$\tan^2 x = \frac{30}{7} \Rightarrow \tan x = \pm\frac{\sqrt{30}}{\sqrt{7}} = \pm\frac{\sqrt{30}\times\sqrt{7}}{7} = \pm\frac{\sqrt{210}}{7}$.

x is **OBTUSE** so $\tan x = -\frac{\sqrt{210}}{7}$ (the result is negative as $\tan x$ is below the x-axis between $\frac{\pi}{2}$ and π).

Ignore $\frac{\pi}{2} < x < \pi$ and the negative sign for the moment and draw a right-angled triangle with this information:



$$\sin x = \frac{\sqrt{210}}{\sqrt{259}} = \frac{\sqrt{210}\times\sqrt{259}}{259} = \frac{\sqrt{7}\times\sqrt{30}\times\sqrt{7}\times\sqrt{37}}{259}$$
$$= \frac{7\sqrt{1110}}{259} = \frac{\sqrt{1110}}{37}.$$

(the result is positive as $\sin x$ is above the x-axis between $\frac{\pi}{2}$ and π).

6 b)

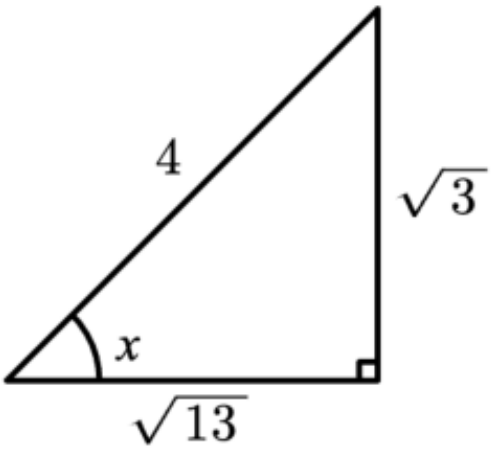
$\cos x = -\frac{7}{\sqrt{259}}$ (the result is negative as $\cos x$ is below the x-axis between $\frac{\pi}{2}$ and π).

7 a)

$\sin^2 x = \frac{3}{16} \Rightarrow \sin x = \pm\frac{\sqrt{3}}{4}$.

$\pi < x < \frac{3\pi}{2}$ so $\sin x = -\frac{\sqrt{3}}{4}$ (the result is negative as $\sin x$ is below the x-axis between π and $\frac{3\pi}{2}$).

Ignore $\pi < x < \frac{3\pi}{2}$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\cos x = -\frac{\sqrt{13}}{4}$ (the result is negative as $\cos x$ is below the x-axis between π and $\frac{3\pi}{2}$).

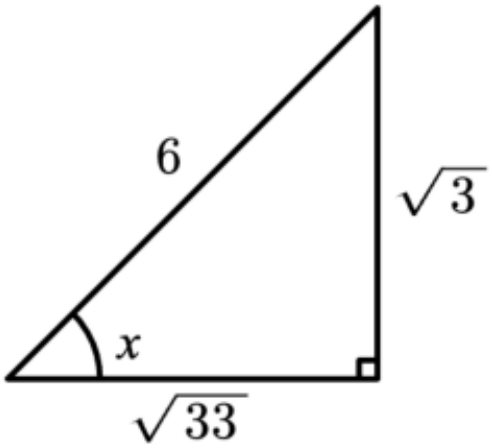
7 b)

$\tan x = \frac{\sqrt{3}}{\sqrt{13}} = \frac{\sqrt{3}\times\sqrt{13}}{13} = \frac{\sqrt{39}}{13}$ (the result is positive as $\tan x$ is above the x-axis between π and $\frac{3\pi}{2}$).

8 a)

$\cos^2 x = \frac{11}{12} \Rightarrow \cos x = \pm\frac{\sqrt{11}}{\sqrt{12}} = \pm\frac{\sqrt{11}}{2\sqrt{3}} = \pm\frac{\sqrt{11}\times\sqrt{3}}{6} = \pm\frac{\sqrt{33}}{6}$.

$\pi < x < \frac{3\pi}{2}$ so $\cos x = -\frac{\sqrt{33}}{6}$ (the result is negative as $\cos x$ is below the x-axis between π and $\frac{3\pi}{2}$). Ignore $\pi < x < \frac{3\pi}{2}$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{\sqrt{3}}{6}$ (the result is negative as $\sin x$ is below the x-axis between π and $\frac{3\pi}{2}$).

8 b)

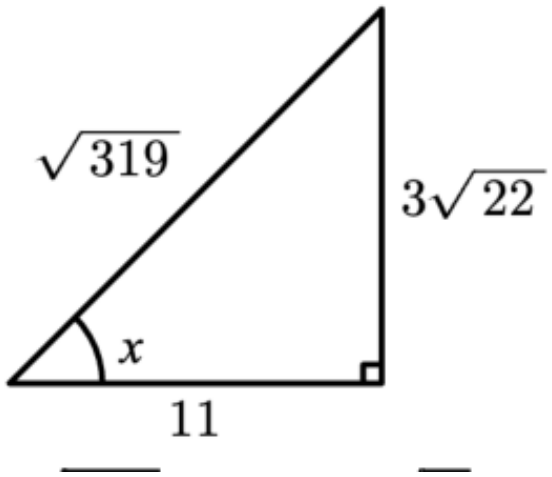
$\tan x = \frac{\sqrt{3}}{\sqrt{33}} = \frac{\sqrt{3}}{\sqrt{3}\times\sqrt{11}} = \frac{1}{\sqrt{11}} = \frac{\sqrt{11}}{11}$ (the result is positive as $\tan x$ is above the x-axis between π and $\frac{3\pi}{2}$).

9 a)

$\tan^2 x = \frac{18}{11} \Rightarrow \tan x = \pm\frac{\sqrt{18}}{\sqrt{11}} = \pm\frac{3\sqrt{2}}{\sqrt{11}} = \pm\frac{3\sqrt{2}\times\sqrt{11}}{11} = \pm\frac{3\sqrt{22}}{11}$.

$\pi < x < \frac{3\pi}{2}$ so $\tan x = \frac{3\sqrt{22}}{11}$ (the result is positive as $\tan x$ is above the x-axis between $\frac{\pi}{2}$ and π).

Ignore $\pi < x < \frac{3\pi}{2}$ for the moment and draw a right-angled triangle with this information:



$$\sin x = -\frac{3\sqrt{22}}{\sqrt{319}} = -\frac{3\sqrt{22}\times\sqrt{319}}{319} = -\frac{3\sqrt{7018}}{319}$$
$$= -\frac{3\times 11\sqrt{58}}{319} = \frac{3\sqrt{58}}{29}.$$

(the result is negative as $\sin x$ is below the x-axis between π and $\frac{3\pi}{2}$).

9 b)

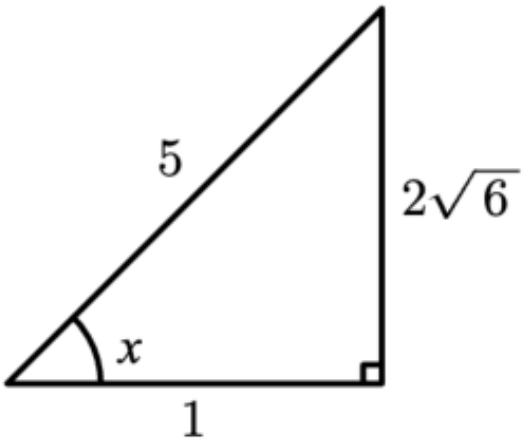
$\cos x = -\frac{11}{\sqrt{319}} = -\frac{11\sqrt{319}}{319} = -\frac{\sqrt{319}}{29}$ (the result is negative as $\cos x$ is below the x-axis between π and $\frac{3\pi}{2}$).

10 a)

$\sin^2 x = \frac{24}{25} \Rightarrow \sin x = \pm\frac{\sqrt{24}}{5} = \pm\frac{2\sqrt{6}}{5}$.

$\frac{3\pi}{2} < x < 2\pi$ so $\sin x = -\frac{2\sqrt{6}}{5}$ (the result is negative as $\sin x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π).

Ignore $\frac{3\pi}{2} < x < 2\pi$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\cos x = \frac{1}{5}$ (the result is positive as $\cos x$ is above the x-axis between $\frac{3\pi}{2}$ and 2π).

10 b)

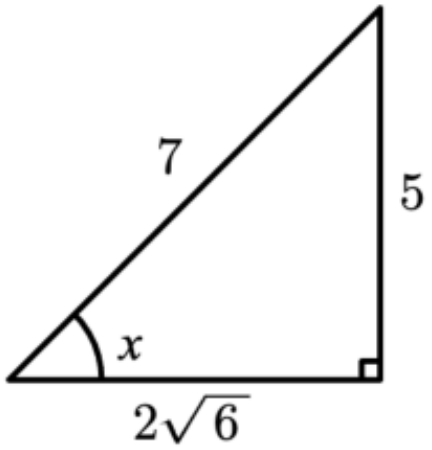
$\tan x = -\frac{2\sqrt{6}}{1} = -2\sqrt{6}$ (the result is negative as $\tan x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π).

11 a)

$\cos^2 x = \frac{24}{49} \Rightarrow \cos x = \pm\frac{\sqrt{24}}{7} = \pm\frac{2\sqrt{6}}{7}$.

$\frac{3\pi}{2} < x < 2\pi$ so $\cos x = \frac{2\sqrt{6}}{7}$ (the result is positive as $\cos x$ is above the x-axis between $\frac{3\pi}{2}$ and 2π).

Ignore $\frac{3\pi}{2} < x < 2\pi$ for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{5}{7}$ (the result is negative as $\sin x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π).

11 b)

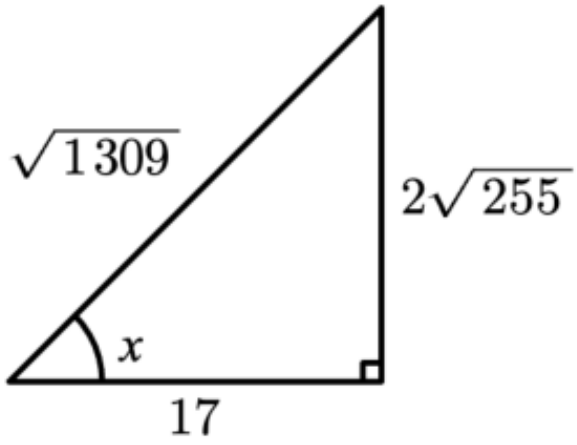
$\tan x = -\frac{5}{2\sqrt{6}} = -\frac{5\sqrt{6}}{12}$ (the result is negative as $\tan x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π).

12 a)

$\tan^2 x = \frac{60}{17} \Rightarrow \tan x = \pm\frac{\sqrt{60}}{\sqrt{17}} = \pm\frac{2\sqrt{15}}{\sqrt{17}} = \pm\frac{2\sqrt{15}\times\sqrt{17}}{17} = \pm\frac{2\sqrt{255}}{17}$.

$\frac{3\pi}{2} < x < 2\pi$ so $\tan x = -\frac{2\sqrt{255}}{17}$ (the result is negative as $\tan x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π).

Ignore $\frac{3\pi}{2} < x < 2\pi$ for the moment and draw a right-angled triangle with this information:



$$\sin x = -\frac{2\sqrt{255}}{\sqrt{1309}} = -\frac{2\sqrt{255}\times\sqrt{1309}}{1309} = -\frac{2\sqrt{17}\times\sqrt{15}\times\sqrt{17}\times\sqrt{77}}{1309}$$

$$= -\frac{34\sqrt{1155}}{1309} = -\frac{2\sqrt{1155}}{77}.$$

(the result is negative as $\sin x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π).

12 b)

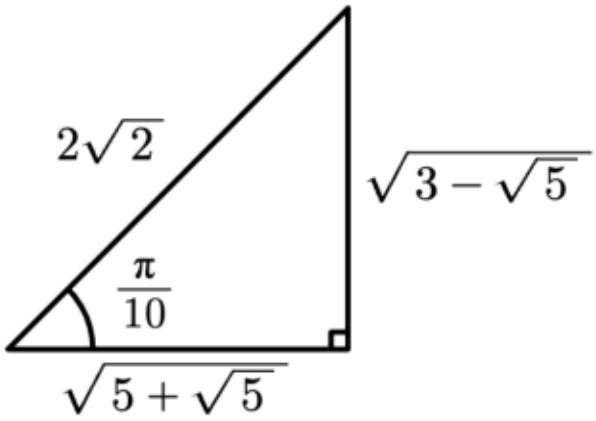
$\cos x = \frac{17}{\sqrt{1309}} = \frac{17\sqrt{1309}}{1309} = \frac{\sqrt{1309}}{77}$ (the result is positive as $\cos x$ is above the x-axis between $\frac{3\pi}{2}$ and 2π).

13 a)

$$\cos\frac{\pi}{10} = \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}} = \sqrt{\frac{5+\sqrt{5}}{8}} = \frac{\sqrt{5+\sqrt{5}}}{\sqrt{8}}$$
$$= \frac{\sqrt{5+\sqrt{5}}}{2\sqrt{2}} = \frac{\text{Adjacent}}{\text{Hypotenuse}}.$$

Opposite $= \sqrt{(2\sqrt{2})^2 - (\sqrt{5+\sqrt{5}})^2}$

$= \sqrt{8 - (5+\sqrt{5})} = \sqrt{3-\sqrt{5}}.$



$\sin\frac{\pi}{10} = \frac{\sqrt{3-\sqrt{5}}}{2\sqrt{2}}$.

13 b)

$$\tan\frac{\pi}{10} = \frac{\sqrt{3-\sqrt{5}}}{\sqrt{5+\sqrt{5}}} = \sqrt{\frac{3-\sqrt{5}}{5+\sqrt{5}}} = \sqrt{\frac{3-\sqrt{5}}{5+\sqrt{5}} \times \frac{5-\sqrt{5}}{5-\sqrt{5}}}$$
$$= \sqrt{\frac{15-8\sqrt{5}+5}{25-5}} = \sqrt{\frac{20-8\sqrt{5}}{20}} = \sqrt{\frac{20}{20} - \frac{8\sqrt{5}}{20}} = \sqrt{1-\frac{2\sqrt{5}}{5}}$$
$$= \sqrt{1-\frac{2\sqrt{5}}{5} \times \frac{\sqrt{5}}{\sqrt{5}}} = \sqrt{1-\frac{10}{5\sqrt{5}}} = \sqrt{1-\frac{2}{\sqrt{5}}}.$$

14)

The first error is going from $3\sin x\cos x = \sin x$ to $3\cos x = 1$. Serkan has divided his equation by $\sin x$, but this will lose the solutions when $\sin x = 0$. He should factorise rearrange and factorise instead:

$$3\sin x\cos x - \sin x = 0 \Rightarrow \sin x(3\cos x - 1) = 0.$$

The second error is that he has solved $\cos x = \frac{1}{3}$ in radians when the question said to solve between $0°$ and $360°$.

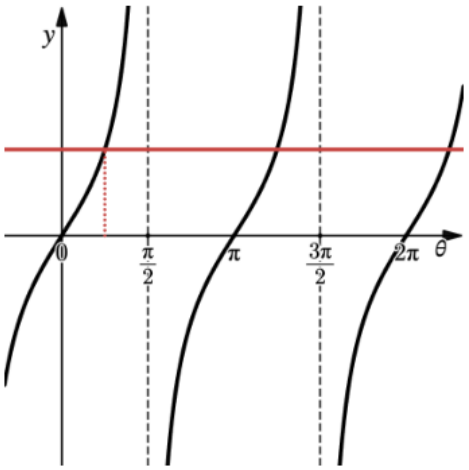
15 a)

$\sin\theta = \cos\theta$ for $0 \leq \theta < 4\pi$.

$\Rightarrow \dfrac{\sin\theta}{\cos\theta} = 1 \Rightarrow \tan\theta = 1.$

Find first value:

$\theta = \arctan 1 = \frac{\pi}{4}$ (in range).



Add period:

$\theta = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$ (in range).

$\theta = \frac{5\pi}{4} + \pi = \frac{9\pi}{4}$ (in range).

$\theta = \frac{9\pi}{4} + \pi = \frac{13\pi}{4}$ (in range).

$\Rightarrow \theta = \{\frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}\}$.

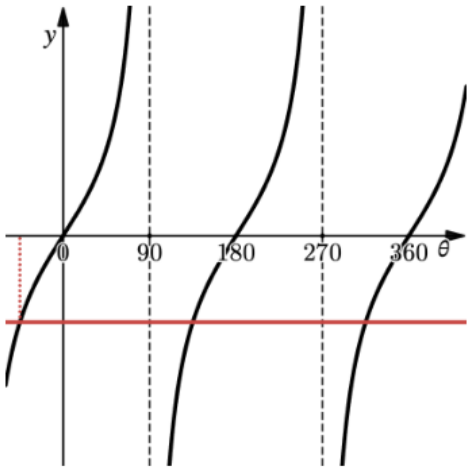
15 b)

$20\sin\theta + 3\cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{20\sin\theta}{\cos\theta} + 3 = 0 \Rightarrow 20\tan\theta + 3 = 0 \Rightarrow \tan\theta = -\frac{3}{20}$.

Find first value:

$\theta = \arctan(-\frac{3}{20}) = -0.1488\ldots$ (not in range).



Add on period:

$\theta = -0.1488\ldots + \pi = 2.9927\ldots$ (in range).

$\theta = 2.9927\ldots + \pi = 6.1342\ldots$ (in range).

$\Rightarrow \theta = \{2.99, 6.13\}$ to 3sf.

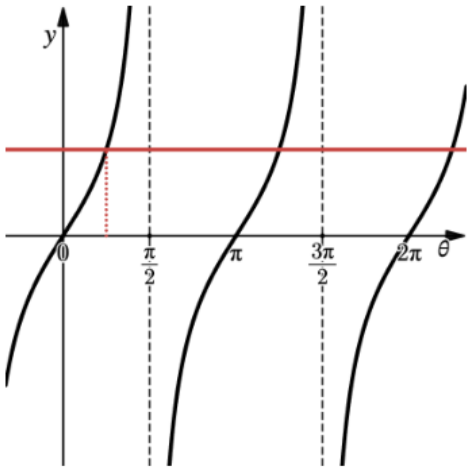
15 c)

$11\cos\theta - 4\sin\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 11 - \dfrac{4\sin\theta}{\cos\theta} = 0 \Rightarrow 11 - 4\tan\theta = 0 \Rightarrow \tan\theta = \frac{11}{4}$.

Find first value:

$\theta = \arctan\frac{11}{4} = 1.2220$ (in range).



Add period:

$\theta = 1.2220 + \pi = 4.3636\ldots$ (in range).

$\Rightarrow \theta = \{1.22, 4.36\}$ to 3sf.

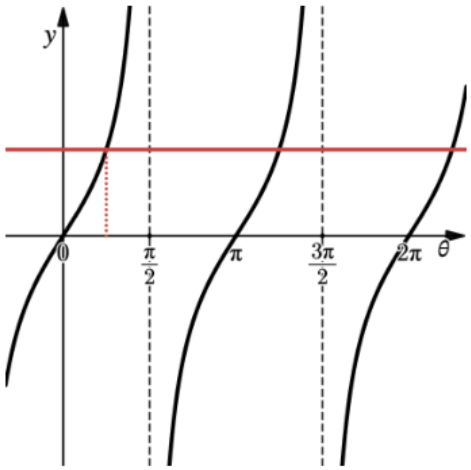
16 a)

$\sin 2\theta = \cos 2\theta$ for $0 \leq \theta < 3\pi$.

$\Rightarrow \dfrac{\sin 2\theta}{\cos 2\theta} = 1 \Rightarrow \tan 2\theta = 1.$

Find first value:

$2\theta = \arctan 1 = \frac{\pi}{4}$.



Apply transformation:

$2\theta = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{8}$ (in range).

Add period:

$\theta = \frac{\pi}{8} + \frac{\pi}{2} = \frac{5\pi}{8}$ (in range).

$\theta = \frac{5\pi}{8} + \frac{\pi}{2} = \frac{9\pi}{8}$ (in range).

$\theta = \frac{9\pi}{8} + \frac{\pi}{2} = \frac{13\pi}{8}$ (in range).

$\theta = \frac{13\pi}{8} + \frac{\pi}{2} = \frac{17\pi}{8}$ (in range).

$\theta = \frac{17\pi}{8} + \frac{\pi}{2} = \frac{21\pi}{8}$ (in range).

$\Rightarrow \theta = \{\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}, \frac{17\pi}{8}, \frac{21\pi}{8}\}$.

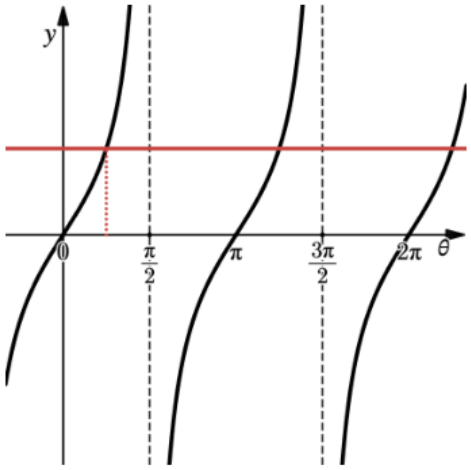
16 b)

$7\sin 4\theta = 2\cos 4\theta$ for $0 \leq \theta < \frac{\pi}{2}$.

$\Rightarrow \dfrac{7\sin 4\theta}{\cos 4\theta} = 2 \Rightarrow 7\tan 4\theta = 2 \Rightarrow \tan 4\theta = \frac{2}{7}$.

Find first value:

$4\theta = \arctan \frac{2}{7} = 0.2782\ldots$



Apply transformation:

$4\theta = 0.2782\ldots \Rightarrow \theta = 0.06957\ldots$ (in range).

Add period:

$\theta = 0.06957\ldots + \frac{\pi}{4} = 0.8549\ldots$ (in range).

$\Rightarrow \theta = \{0.0696, 0.855\}$ to 3sf.

16 c)

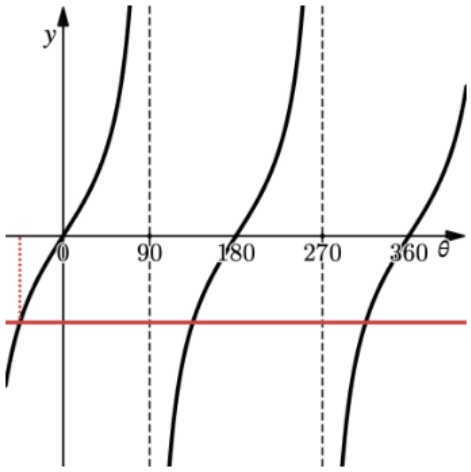
$10\sin\frac{1}{2}\theta + 3\cos\frac{1}{2}\theta = 0$ for $0 \leq \theta < 5\pi$.

$\Rightarrow \dfrac{10\sin\frac{1}{2}\theta}{\cos\frac{1}{2}\theta} + 3 = 0 \Rightarrow 10\tan\frac{1}{2}\theta + 3 = 0$

$\Rightarrow \tan\frac{1}{2}\theta = -\frac{3}{10}$.

Find first value:

$\frac{1}{2}\theta = \arctan(-\frac{3}{10}) = -0.2914\ldots$



Apply transformation:

$\frac{1}{2}\theta = -0.2914\ldots \Rightarrow \theta = -0.5829\ldots$ (not in range).

Add on period:

$\theta = -0.5829\ldots + 2\pi = 5.7002\ldots$ (in range).

$\theta = 5.7002\ldots + 2\pi = 11.9834\ldots$ (in range).

$\Rightarrow \theta = \{5.70, 12.0\}$ to 3sf.

16 d)

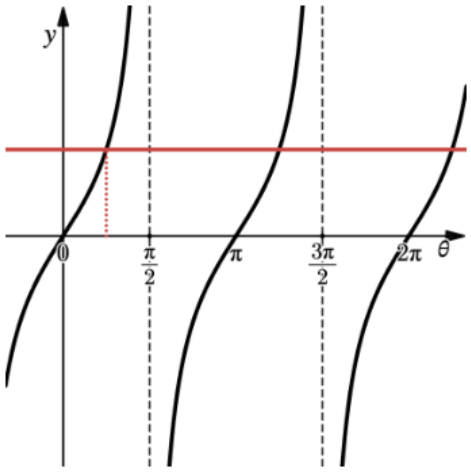
$19\cos\frac{1}{4}\theta - 16\sin\frac{1}{4}\theta = 0$ for $0 \leq \theta < 20\pi$.

$\Rightarrow 19 - \dfrac{16\sin\frac{1}{4}\theta}{\cos\frac{1}{4}\theta} = 0 \Rightarrow 19 - 16\tan\frac{1}{4}\theta = 0$

$\Rightarrow \tan\frac{1}{4}\theta = \frac{19}{16}$.

Find first value:

$\frac{1}{4}\theta = \arctan\frac{19}{16} = 0.8709\ldots$



Apply transformation:

$\frac{1}{4}\theta = 0.8709\ldots \Rightarrow \theta = 3.4836\ldots$ (in range).

Add period:

$\theta = 3.4836\ldots + 4\pi = 16.0499\ldots$ (in range).

$\theta = 16.0499\ldots + 4\pi = 28.6163\ldots$ (in range).

$\theta = 28.6163\ldots + 4\pi = 41.1827\ldots$ (in range).

$\theta = 41.1827\ldots + 4\pi = 53.7490\ldots$ (in range).

$\Rightarrow \theta = \{3.48, 16.0, 28.6, 41.2, 53.7\}$ to 3sf.

17 a)

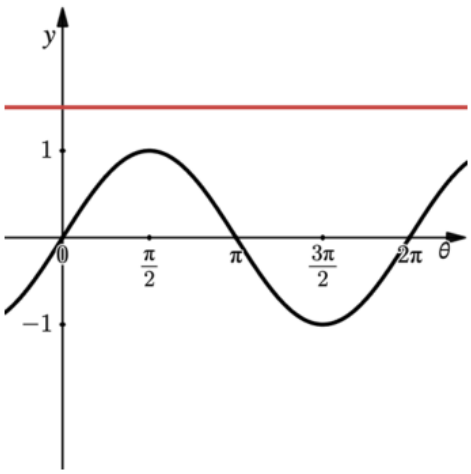
$6\tan\theta + \cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 6 \times \dfrac{\sin\theta}{\cos\theta} + \cos\theta = 0 \Rightarrow 6\sin\theta + \cos^2\theta = 0$

$\Rightarrow 6\sin\theta + (1 - \sin^2\theta) = 0$

$\Rightarrow \sin^2\theta - 6\sin\theta - 1 = 0 \Rightarrow \sin\theta = 3 \pm \sqrt{10}$.

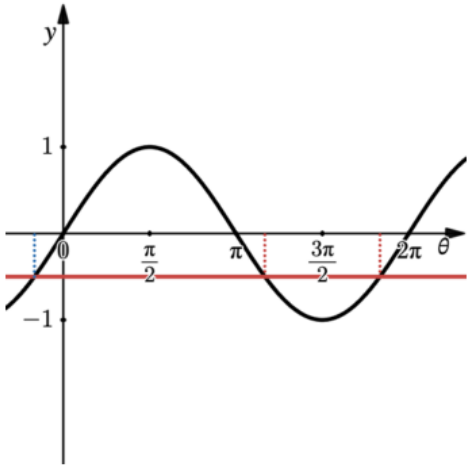
As for $\sin\theta = 3 + \sqrt{10}$ there are no solutions:



Now solve $\sin\theta = 3 - \sqrt{10}$ for $0 \leq \theta < 2\pi$.

Find first value:

$\theta = \arcsin(3 - \sqrt{10}) = -0.1629\ldots$ (not in range).



Find first value between 0 and 2π:

$\theta = \pi + 0.1629\ldots = 3.3045\ldots$ (in range).

Find second value between 0 and 2π:

$\theta = 2\pi - 0.1629\ldots = 6.1201\ldots$ (in range).

$\Rightarrow \theta = \{3.30, 6.12\}$.

17 b)

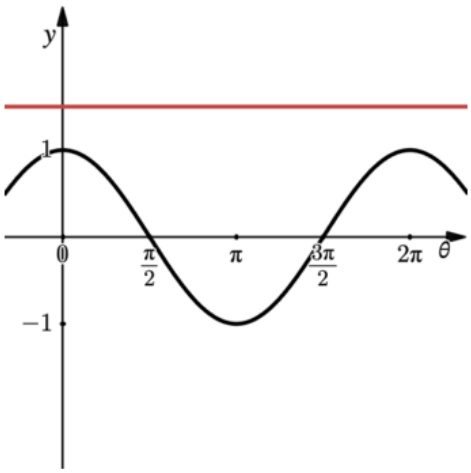
$\sin\theta\tan\theta + 16 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \sin\theta \times \dfrac{\sin\theta}{\cos\theta} + 16 = 0 \Rightarrow \sin^2\theta + 16\cos\theta = 0$

$\Rightarrow (1 - \cos^2\theta) + 16\cos\theta = 0$

$\Rightarrow \cos^2\theta - 16\cos\theta - 1 = 0 \Rightarrow \cos\theta = 8 \pm \sqrt{65}$.

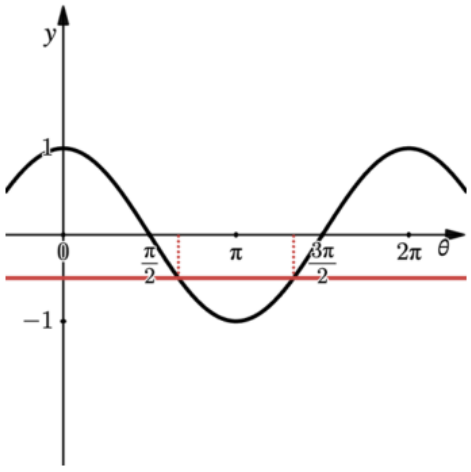
As for $\cos\theta = 8 + 3\sqrt{65}$ there are no solutions:



Now solve $\cos\theta = 8 - \sqrt{65}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos(8 - \sqrt{65}) = 1.6330\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 1.6330\ldots = 4.6500\ldots$

$\Rightarrow \theta = \{1.63, 4.65\}$.

17 c)

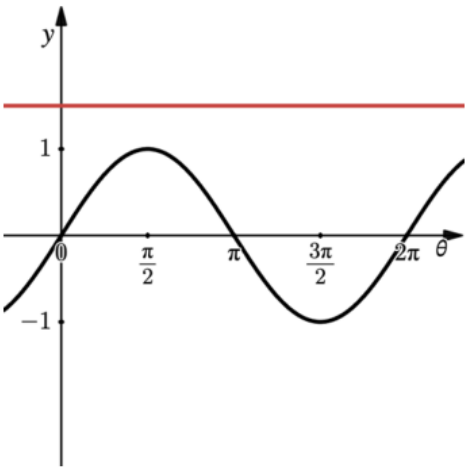
$12\tan\frac{1}{2}\theta + \cos\frac{1}{2}\theta = 0$ for $0 \leq \theta \leq 8\pi$.

$\Rightarrow \dfrac{12\sin\frac{1}{2}\theta}{\cos\frac{1}{2}\theta} + \cos\frac{1}{2}\theta = 0 \Rightarrow 12\sin\frac{1}{2}\theta + \cos^2\frac{1}{2}\theta = 0$

$\Rightarrow 12\sin\frac{1}{2}\theta + (1 - \sin^2\frac{1}{2}\theta) = 0$

$\Rightarrow \sin^2\frac{1}{2}\theta - 12\sin\frac{1}{2}\theta - 1 = 0 \Rightarrow \sin\frac{1}{2}\theta = 6 \pm \sqrt{37}$.

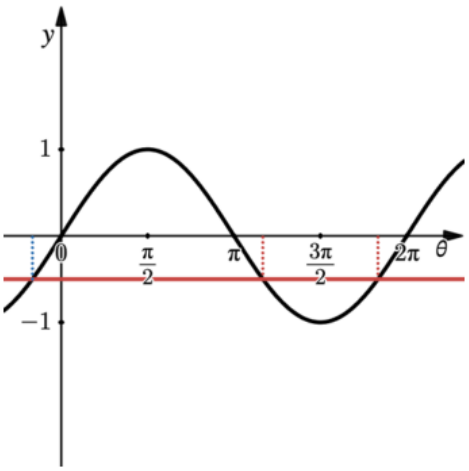
As for $\sin\frac{1}{2}\theta = 6 + \sqrt{37}$ there are no solutions:



Now solve $\sin\frac{1}{2}\theta = 6 - \sqrt{37}$ for $0 \leq \theta < 8\pi$.

Find first value:

$\frac{1}{2}\theta = \arcsin(6 - \sqrt{37}) = -0.08285\ldots$



Find first value between 0 and 2π:

$\frac{1}{2}\theta = \pi + 0.08285\ldots = 3.2244\ldots$

Find second value between 0 and 2π:

$\frac{1}{2}\theta = 2\pi - 0.08285\ldots = 6.2003\ldots$

Apply transformation: $\frac{1}{2}\theta = 3.2244\ldots \Rightarrow \theta = 6.4488\ldots$ (in range).

$\frac{1}{2}\theta = 6.2003\ldots \Rightarrow \theta = 12.4006\ldots$ (in range).

Add period:

$\theta = 6.4488\ldots + 4\pi = 19.0152\ldots$ (in range).

$\theta = 12.4006\ldots + 4\pi = 24.9670\ldots$ (in range).

$\Rightarrow \theta = \{6.45, 12.4, 19.0, 25.0\}$.

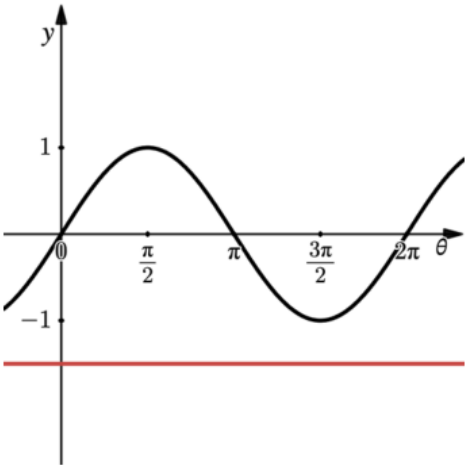
18 a)

$3\cos^2\theta = 8\sin\theta$ for $0 \leq \theta < \pi$.

$\Rightarrow 3(1 - \sin^2\theta) = 8\sin\theta \Rightarrow 3 - 3\sin^2\theta = 8\sin\theta$

$\Rightarrow 3\sin^2\theta + 8\sin\theta - 3 = 0 \Rightarrow \sin\theta = -3$ or $\sin\theta = \frac{1}{3}$.

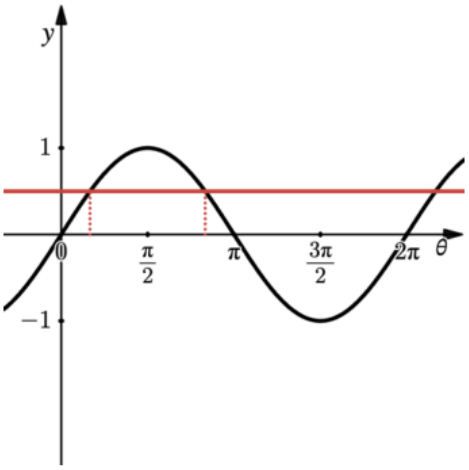
As for $\sin\theta = -3$ there are no solutions:



Now solve $\sin\theta = \frac{1}{3}$ for $0 \leq \theta < \pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{1}{3} = 0.3398\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = \pi - 0.3398\ldots = 2.8017\ldots$

$\Rightarrow \theta = \{0.340, 2.80\}$ to 3sf.

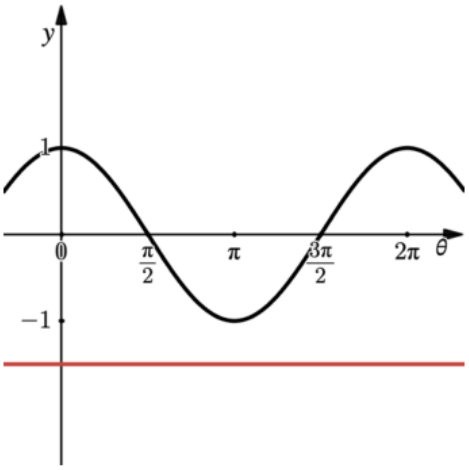
18 b)

$9\sin^2\theta = 80\cos\theta$ for $0 \leq \theta < 3\pi$.

$\Rightarrow 9(1-\cos^2\theta) = 80\cos\theta \Rightarrow 9 - 9\cos^2\theta = 80\cos\theta$

$\Rightarrow 9\cos^2\theta + 80\cos\theta - 9 = 0 \Rightarrow \cos\theta = -9$ or $\cos\theta = \frac{1}{9}$.

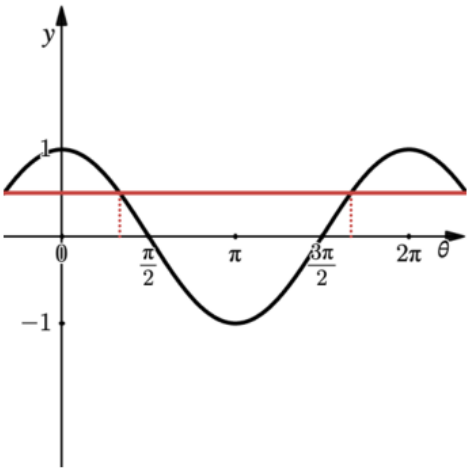
As for $\cos\theta = -9$ there are no solutions:



Now solve $\cos\theta = \frac{1}{9}$ for $0 \leq \theta < 3\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{1}{9} = 1.4594\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 1.4594\ldots = 4.8237\ldots$

Add period:

$\theta = 1.4594\ldots + 2\pi = 7.7426\ldots$

$\Rightarrow \theta = \{1.46, 4.82, 7.74\}$ to 3sf.

18 c)

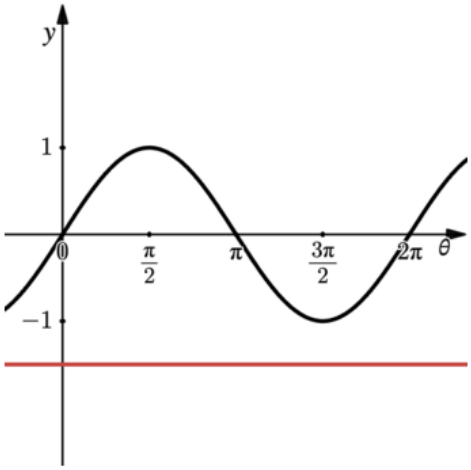
$12\cos^2\theta = 143\sin\theta$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 12(1-\sin^2\theta) = 143\sin\theta$

$\Rightarrow 12 - 12\sin^2\theta = 143\sin\theta$

$\Rightarrow 12\sin^2\theta + 143\sin\theta - 12 = 0 \Rightarrow \sin\theta = -12$ or $\sin\theta = \frac{1}{12}$.

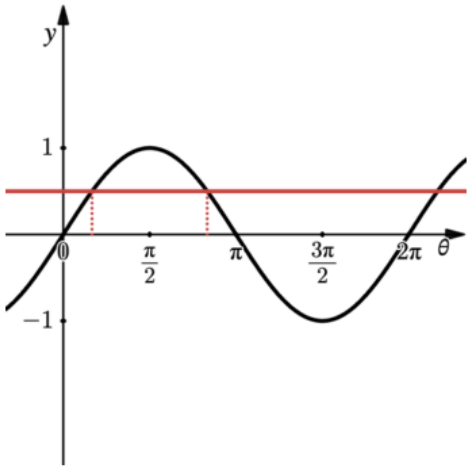
As for $\sin\theta = -12$ there are no solutions:



Now solve $\sin\theta = \frac{1}{12}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{1}{12} = 0.08343\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = \pi - 0.08343\ldots = 3.0581\ldots$

$\Rightarrow \theta = \{0.0834, 3.06\}$ to 3sf.

18 d)

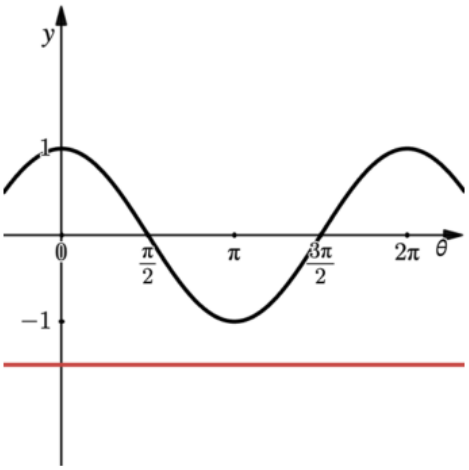
$50\sin^2\theta = 2499\cos\theta$ for $-\pi \leq \theta < \pi$.

$\Rightarrow 50(1-\cos^2\theta) = 2499\cos\theta$

$\Rightarrow 50 - 50\cos^2\theta = 2499\cos\theta$

$\Rightarrow 50\cos^2\theta + 2499\cos\theta - 50 = 0 \Rightarrow \cos\theta = -50$ or $\cos\theta = \frac{1}{50}$.

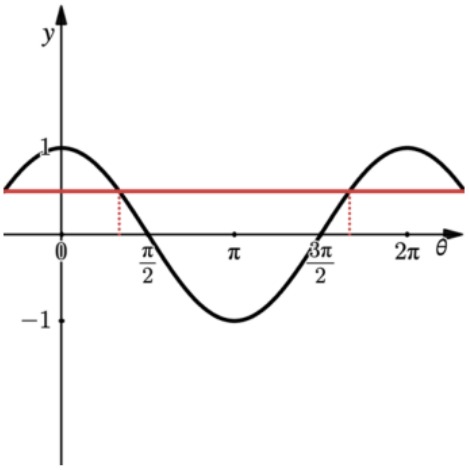
As for $\cos\theta = -50$ there are no solutions:



Now solve $\cos\theta = \frac{1}{50}$ for $-\pi \leq \theta < \pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{1}{50} = 1.5507\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 1.5507\ldots = 4.7323\ldots$ (not in range).

Subtract period:

$\theta = 4.7323\ldots - 2\pi = -1.5507\ldots$ (in range).

$\Rightarrow \theta = \{-1.55, 1.55\}$ to 3sf.

19 a)

$4\cos^2\theta + 9\sin\theta - 6 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 4(1 - \sin^2\theta) + 9\sin\theta - 6 = 0$

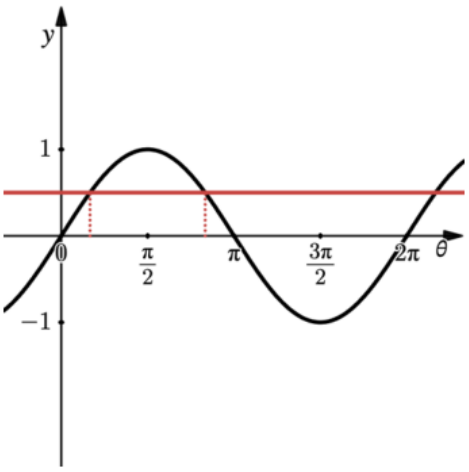
$\Rightarrow 4 - 4\sin^2\theta + 9\sin\theta - 6 = 0$

$\Rightarrow 4\sin^2\theta - 9\sin\theta + 2 = 0 \Rightarrow \sin\theta = \frac{1}{4}$ or $\sin\theta = 2$.

First solve $\sin\theta = \frac{1}{4}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

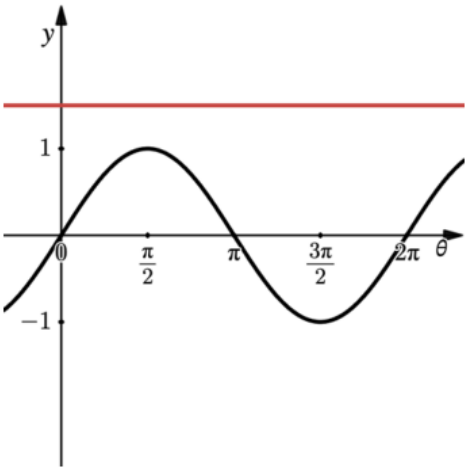
$\theta = \arcsin\frac{1}{4} = 0.2526\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = \pi - 0.2526\ldots = 2.8889\ldots$

As for $\sin\theta = 2$ there are no solutions:



$\Rightarrow \theta = \{0.253, 2.89\}$ to 3sf.

19 b)

$9\sin^2\theta + 3\cos\theta - 7 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 9(1 - \cos^2\theta) + 3\cos\theta - 7 = 0$

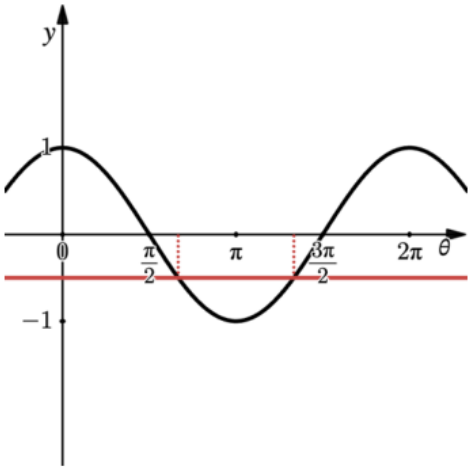
$\Rightarrow 9 - 9\cos^2\theta + 3\cos\theta - 7 = 0$

$\Rightarrow 9\cos^2\theta - 3\cos\theta - 2 = 0 \Rightarrow \cos\theta = -\frac{1}{3}$ or $\cos\theta = \frac{2}{3}$.

First solve $\cos\theta = -\frac{1}{3}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos(-\frac{1}{3}) = 1.9106\ldots$ (in range).



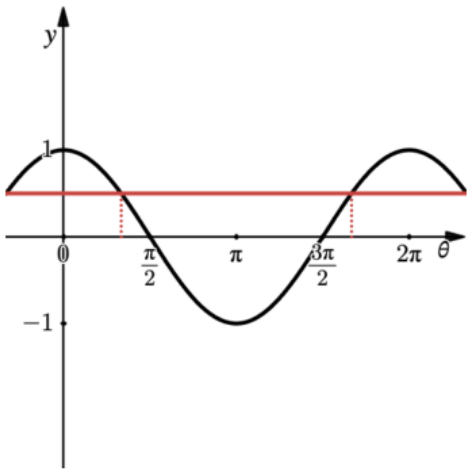
Find second value between 0 and 2π:

$\theta = 2\pi - 1.9106\ldots = 4.3725\ldots$ (in range).

Now solve $\cos\theta = \frac{2}{3}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{2}{3} = 0.8410\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 0.8410\ldots = 5.4421\ldots$ (in range).

$\Rightarrow \theta = \{0.841, 1.91, 4.37, 5.44\}$ to 3sf.

19 c)

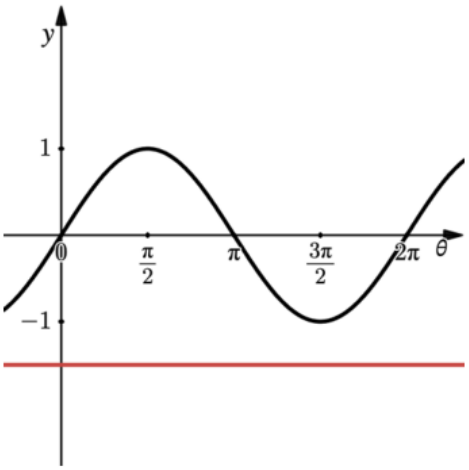
$15\cos^2\theta - 31\sin\theta - 25 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 15(1 - \sin^2\theta) - 31\sin\theta - 25 = 0$

$\Rightarrow 15 - 15\sin^2\theta - 31\sin\theta - 25 = 0$

$\Rightarrow 15\sin^2\theta + 31\sin\theta + 10 = 0 \Rightarrow \sin\theta = -\frac{5}{3}$ or $\sin\theta = -\frac{2}{5}$.

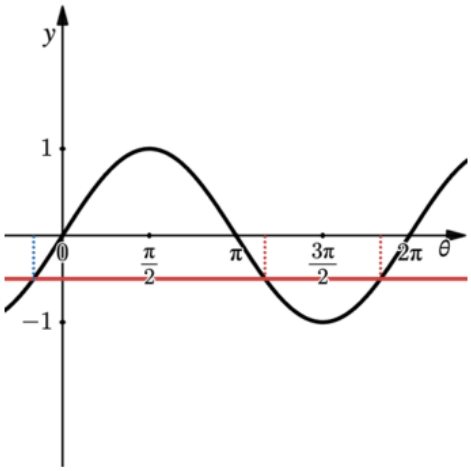
As for $\sin\theta = -\frac{5}{3}$ there are no solutions:



Now solve $\sin\theta = -\frac{2}{5}$ for $0 \leq \theta < 2\pi$.

Find first value:

$\theta = \arcsin(-\frac{2}{5}) = -0.4115\ldots$ (not in range).



Find first value between 0 and 2π:

$\theta = \pi + 0.4115\ldots = 3.5531\ldots$ (in range).

Find second value between 0 and 2π:

$\theta = 2\pi - 0.4115\ldots = 5.8716\ldots$ (in range).

$\Rightarrow x = \{3.55, 5.87\}$ to 3sf.

19 d)

$27\sin^2\theta + 12\cos\theta - 7 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 27(1 - \cos^2\theta) + 12\cos\theta - 7 = 0$

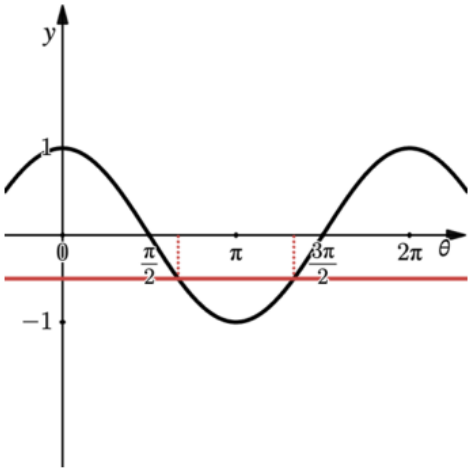
$\Rightarrow 27 - 27\cos^2\theta + 12\cos\theta - 7 = 0$

$\Rightarrow 27\cos^2\theta - 12\cos\theta - 20 = 0 \Rightarrow \cos\theta = -\frac{2}{3}$ or $\cos\theta = \frac{10}{9}$.

First solve $\cos\theta = -\frac{2}{3}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

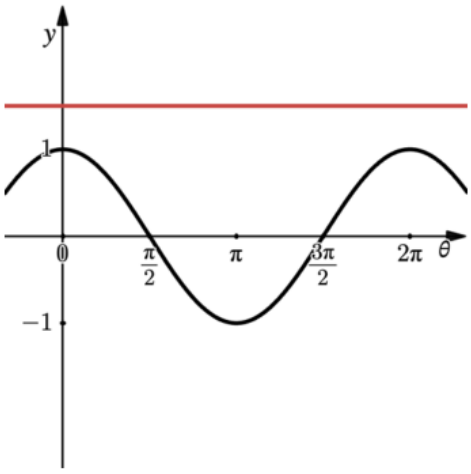
$\theta = \arccos(-\frac{2}{3}) = 2.3005\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 2.3005\ldots = 3.9826\ldots$ (in range).

As for $\cos\theta = \frac{10}{9}$ there are no solutions:



$\theta = \{2.30, 3.98\}$.

20 a)

$8\sin^2\theta - 26\sin\theta - 16\cos^2\theta + 21 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 8\sin^2\theta - 26\sin\theta - 16(1 - \sin^2\theta) + 21 = 0$

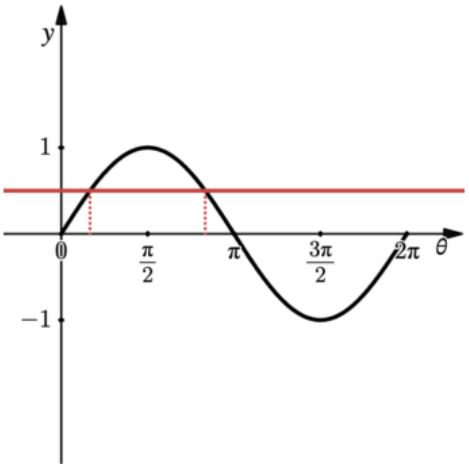
$\Rightarrow 8\sin^2\theta - 26\sin\theta - 16 + 16\sin^2\theta + 21 = 0$

$\Rightarrow 24\sin^2\theta - 26\sin\theta + 5 = 0 \Rightarrow \sin\theta = \frac{1}{4}$ or $\sin\theta = \frac{5}{6}$.

First solve $\sin\theta = \frac{1}{4}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{1}{4} = 0.2526\ldots$ (in range).



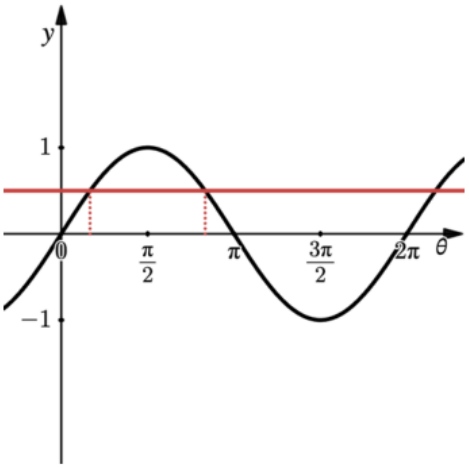
Find second value between 0 and 2π:

$\theta = \pi - 0.2526\ldots = 2.8889\ldots$

Now solve $\sin\theta = \frac{5}{6}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{5}{6} = 0.9851\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = \pi - 0.9851\ldots = 2.1564\ldots$

$\theta = \{0.253, 0.985, 2.16, 2.89\}$.

20 b)

$13\cos^2\theta + 13\cos\theta - 22\sin^2\theta + 10 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 13\cos^2\theta + 13\cos\theta - 22(1 - \cos^2\theta) + 10 = 0$

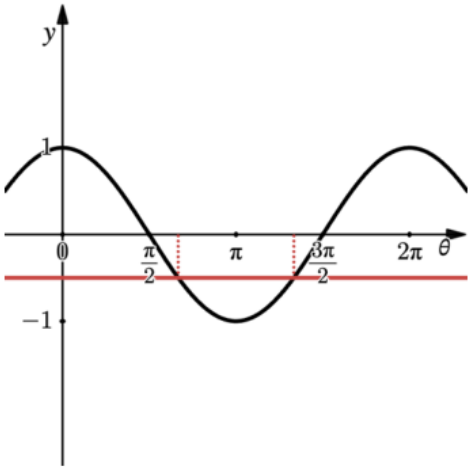
$\Rightarrow 13\cos^2\theta + 13\cos\theta - 22 + 22\cos^2\theta + 10 = 0$

$\Rightarrow 35\cos^2\theta + 13\cos\theta - 12 = 0 \Rightarrow \cos\theta = -\frac{4}{5}$ or $\cos\theta = \frac{3}{7}$.

First solve $\cos\theta = -\frac{4}{5}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos(-\frac{4}{5}) = 2.4980\ldots$ (in range).



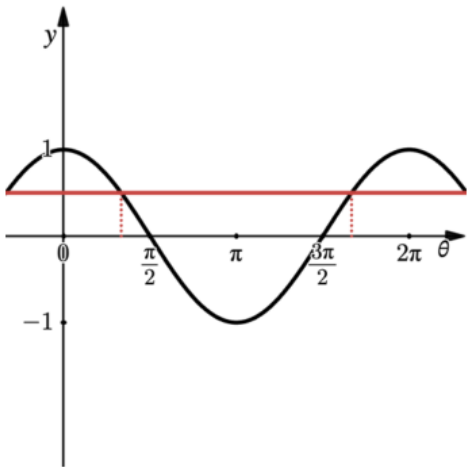
Find second value between 0 and 2π:

$\theta = 2\pi - 2.4980\ldots = 3.7850\ldots$ (in range).

Now solve $\cos\theta = \frac{3}{7}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{3}{7} = 1.1278\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 1.1278\ldots = 5.1553\ldots$ (in range).

$\Rightarrow \theta = \{1.13, 2.50, 3.79, 5.16\}$ to 3sf.

20 c)

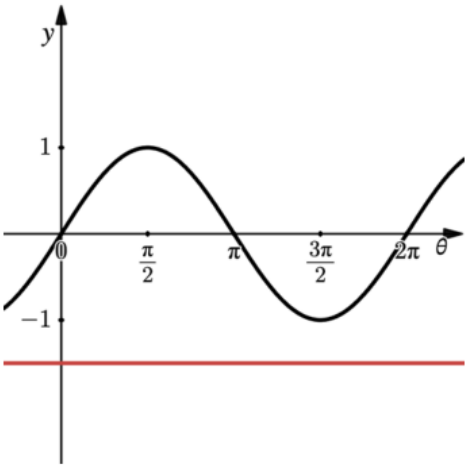
$20\sin^2\theta - 4\cos^2\theta + 38\sin\theta - 51 = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 20\sin^2\theta - 4(1 - \sin^2\theta) + 38\sin\theta - 51 = 0$

$\Rightarrow 20\sin^2\theta - 4 + 4\sin^2\theta + 38\sin\theta - 51 = 0$

$\Rightarrow 24\sin^2\theta + 38\sin\theta - 55 = 0 \Rightarrow \sin\theta = -\frac{5}{2}$ or $\sin\theta = \frac{11}{12}$.

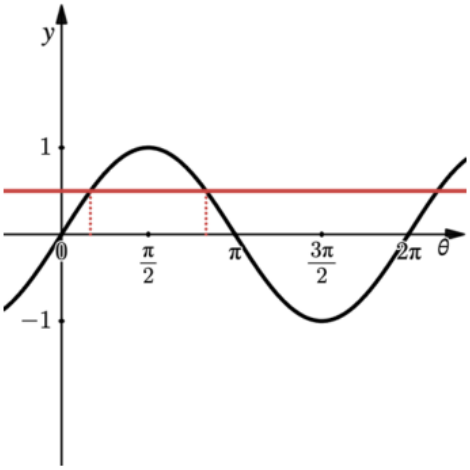
As for $\sin\theta = -\frac{5}{2}$ there are no solutions:



Now solve $\sin\theta = \frac{11}{12}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{11}{12} = 1.1596\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = \pi - 1.1596\ldots = 1.9819\ldots$

$\theta = \{1.16, 1.98\}$ to 3sf.

20 d)

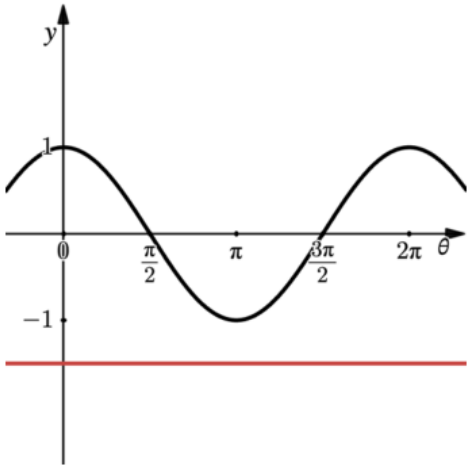
$50\sin^2\theta + 28 = 30\cos^2\theta + 31\cos\theta$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 50(1 - \cos^2\theta) + 28 = 30\cos^2\theta + 31\cos\theta$

$\Rightarrow 50 - 50\cos^2\theta + 28 = 30\cos^2\theta + 31\cos\theta$

$\Rightarrow 80\cos^2\theta + 31\cos\theta - 78 = 0 \Rightarrow \cos\theta = -\frac{6}{5}$ or $\cos\theta = \frac{13}{16}$.

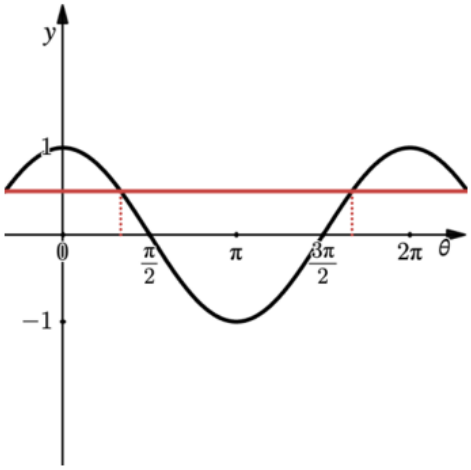
As for $\cos\theta = -\frac{6}{5}$ there are no solutions:



Now solve $\cos\theta = \frac{13}{16}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{13}{16} = 0.6223\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 0.6223\ldots = 5.6608\ldots$ (in range).

$\Rightarrow \theta = \{0.622, 5.66\}$ to 3sf.

21 a)

$20\sin^2 x - 14\sin x - 1 = 4\cos^2 x$ for $-2\pi \leq x < 2\pi$.

$\Rightarrow 20\sin^2 x - 14\sin x - 1 = 4(1 - \sin^2 x)$

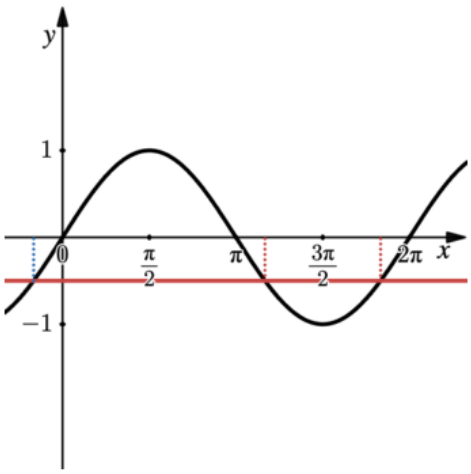
$\Rightarrow 20\sin^2 x - 14\sin x - 1 = 4 - 4\sin^2 x$

$\Rightarrow 24\sin^2 x - 14\sin x - 5 = 0 \Rightarrow \sin x = -\frac{1}{4}$ or $\sin x = \frac{5}{6}$.

First solve $\sin x = -\frac{1}{4}$ for $-2\pi \leq x < 2\pi$.

Find first value:

$x = \arcsin(-\frac{1}{4}) = -0.2526\ldots$ (in range).



Find first value between 0 and 2π:

$x = \pi + 0.2526\ldots = 3.3942\ldots$ (in range).

Find second value between 0 and 2π:

$x = 2\pi - 0.2526\ldots = 6.0305\ldots$ (in range).

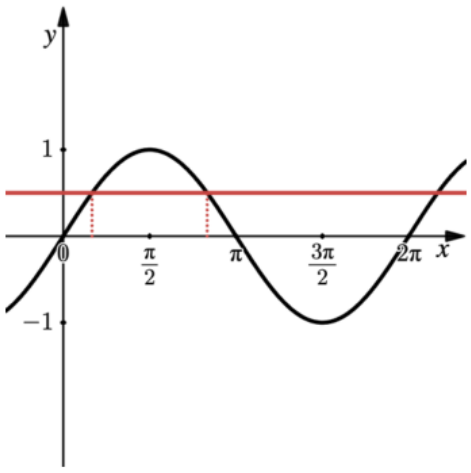
Subtract period:

$x = 3.3942\ldots - 2\pi = -2.8889\ldots$

Now solve $\sin x = \frac{5}{6}$ for $-2\pi \leq x < 2\pi$.

Find first value between 0 and 2π:

$x = \arcsin\frac{5}{6} = 0.9851\ldots$ (in range).



Find second value between 0 and 2π:

$x = \pi - 0.9851\ldots = 2.1564\ldots$ (in range).

Subtract period:

$x = 0.9851\ldots - 2\pi = -5.2980\ldots$ (in range).

$x = 2.1564\ldots - 2\pi = -4.1267\ldots$ (in range).

$\Rightarrow x = \{-5.30, -4.13, -2.89, -0.253, 0.985, 2.16, 3.39, 6.03\}$ to 3sf.

21 b)

$3\theta + \frac{\pi}{3} = 2.1564\ldots \Rightarrow \theta = 0.3697\ldots = 0.370$ to 3sf.

22 a)

$36\sin\theta + \dfrac{10}{\cos\theta} = 24\tan\theta + 15$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 36\sin\theta + \dfrac{10}{\cos\theta} = \dfrac{24\sin\theta}{\cos\theta} + 15$

$\Rightarrow 36\sin\theta\cos\theta + 10 = 24\sin\theta + 15\cos\theta$

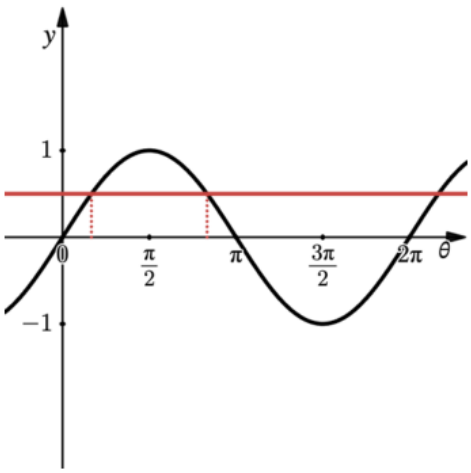
$\Rightarrow 36\sin\theta\cos\theta - 24\sin\theta - 15\cos\theta + 10 = 0$

$\Rightarrow (12\sin\theta - 5)(3\cos\theta - 2) = 0 \Rightarrow \sin\theta = \frac{5}{12}$ or $\cos\theta = \frac{2}{3}$.

First solve $\sin\theta = \frac{5}{12}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{5}{12} = 0.4297\ldots$ (in range).



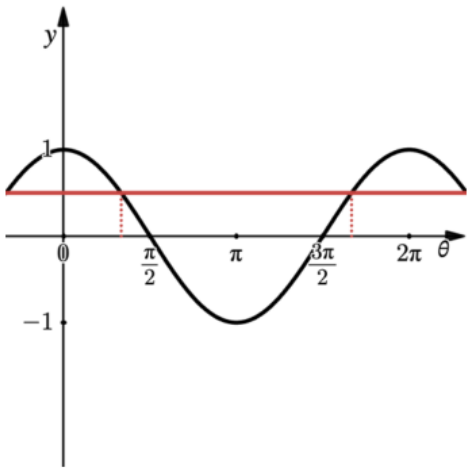
Find second value between 0 and 2π:

$\theta = \pi - 0.4297\ldots = 2.7118\ldots$ (in range).

Now solve $\cos\theta = \frac{2}{3}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{2}{3} = 0.8410\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 0.8410\ldots = 5.4421\ldots$ (in range).

$\Rightarrow \theta = \{0.430, 0.841, 2.71, 5.44\}$ to 3sf.

22 b)

$3\sin\theta\tan\theta = 21\cos\theta + 14$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 3\sin\theta \times \dfrac{\sin\theta}{\cos\theta} = 21\cos\theta + 14$

$\Rightarrow 3\sin^2\theta = 21\cos^2\theta + 14\cos\theta$

$\Rightarrow 3(1 - \cos^2\theta) = 21\cos^2\theta + 14\cos\theta$

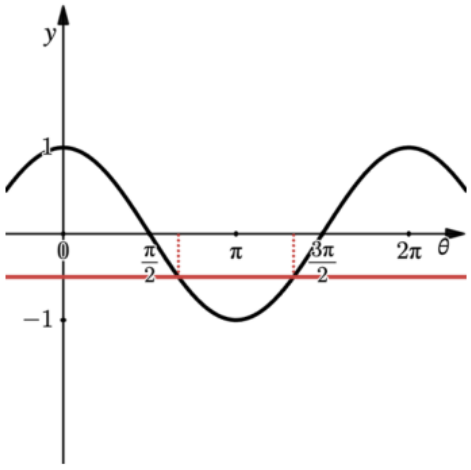
$\Rightarrow 3 - 3\cos^2\theta = 21\cos^2\theta + 14\cos\theta$

$\Rightarrow 24\cos^2\theta + 14\cos\theta - 3 = 0 \Rightarrow \cos\theta = -\frac{3}{4}$ or $\cos\theta = \frac{1}{6}$.

First solve $\cos\theta = -\frac{3}{4}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos(-\frac{3}{4}) = 2.4188\ldots$ (in range).



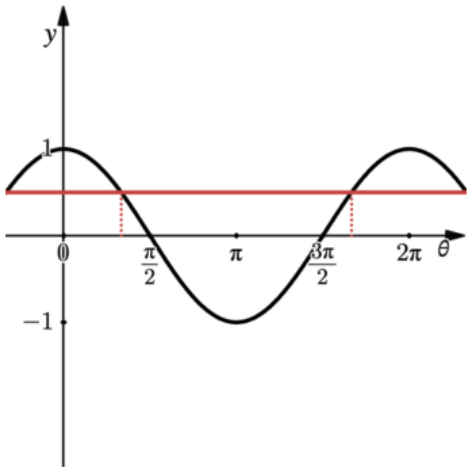
Find second value between 0 and 2π:

$\theta = 2\pi - 2.4188\ldots = 3.8643\ldots$ (in range).

Now solve $\cos\theta = \frac{1}{6}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{1}{6} = 1.4033\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 1.4033\ldots = 4.8798\ldots$ (in range).

$\Rightarrow \theta = \{1.40, 2.42, 3.86, 4.88\}$ to 3sf.

22 c)

$\dfrac{6\cos\theta}{\tan\theta} = 29\sin\theta - 1$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{6\cos\theta}{\frac{\sin\theta}{\cos\theta}} = 29\sin\theta - 1$

$\Rightarrow 6\cos\theta \times \dfrac{\cos\theta}{\sin\theta} = 29\sin\theta - 1$

$\Rightarrow 6\cos^2\theta = 29\sin^2\theta - \sin\theta$

$\Rightarrow 6(1 - \sin^2\theta) = 29\sin^2\theta - \sin\theta$

$\Rightarrow 6 - 6\sin^2\theta = 29\sin^2\theta - \sin\theta$

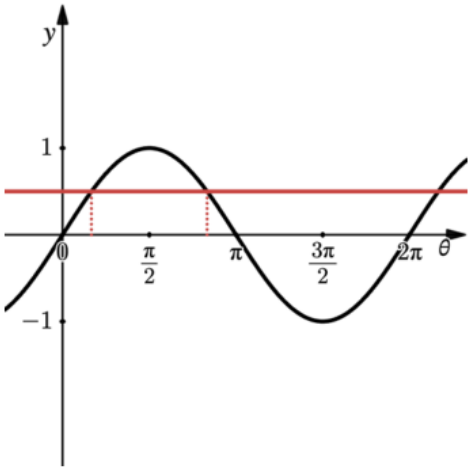
$\Rightarrow 0 = 35\sin^2\theta - \sin\theta - 6 = 0$

$\Rightarrow \sin\theta = \frac{3}{7}$ or $\sin\theta = -\frac{2}{5}$.

First solve $\sin\theta = \frac{3}{7}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arcsin\frac{3}{7} = 0.4429\ldots$ (in range).



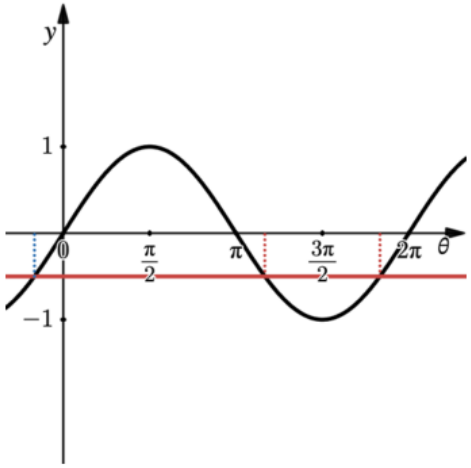
Find second value between 0 and 2π:

$\theta = \pi - 0.4429\ldots = 2.6986\ldots$ (in range).

Now solve $\sin\theta = -\frac{2}{5}$ for $0 \leq \theta < 2\pi$.

Find first value:

$\theta = \arcsin(-\frac{2}{5}) = -0.4115\ldots$ (not in range).



Find first value between 0 and 2π:

$\theta = \pi + 0.4115\ldots = 3.5531\ldots$ (in range).

Find second value between 0 and 2π:

$\theta = 2\pi - 0.4115\ldots = 5.8716\ldots$ (in range).

$\Rightarrow \theta = \{0.443, 2.70, 3.55, 5.87\}$ to 3sf.

22 d)

$10\sin\theta\cos\theta - 2\cos^2\theta + 5\cos\theta = 2\sin\theta(2 + \sin\theta)$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 10\sin\theta\cos\theta - 2\cos^2\theta + 5\cos\theta = 4\sin\theta + 2\sin^2\theta$

$\Rightarrow 10\sin\theta\cos\theta + 5\cos\theta = 4\sin\theta + 2\sin^2\theta + 2\cos^2\theta$

$\Rightarrow 10\sin\theta\cos\theta + 5\cos\theta = 4\sin\theta + 2(\sin^2\theta + \cos^2\theta)$

$\Rightarrow 10\sin\theta\cos\theta + 5\cos\theta = 4\sin\theta + 2$

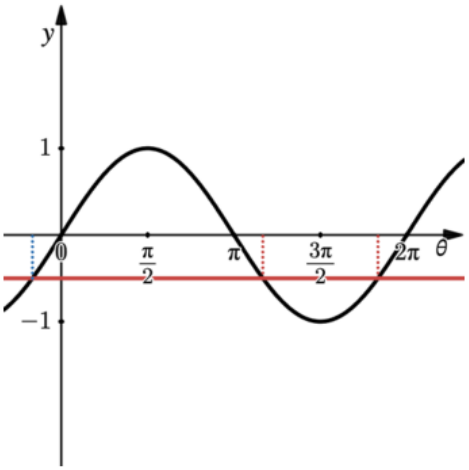
$\Rightarrow 10\sin\theta\cos\theta + 5\cos\theta - 4\sin\theta - 2 = 0$

$\Rightarrow (2\sin\theta + 1)(5\cos\theta - 2) = 0 \Rightarrow \sin\theta = -\frac{1}{2}$ or $\cos\theta = \frac{2}{5}$.

Now solve $\sin\theta = -\frac{1}{2}$ for $0 \leq \theta < 2\pi$.

Find first value:

$\theta = \arcsin(-\frac{1}{2}) = -\frac{\pi}{6}$ (not in range).



Find first value between 0 and 2π:

$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$ (in range).

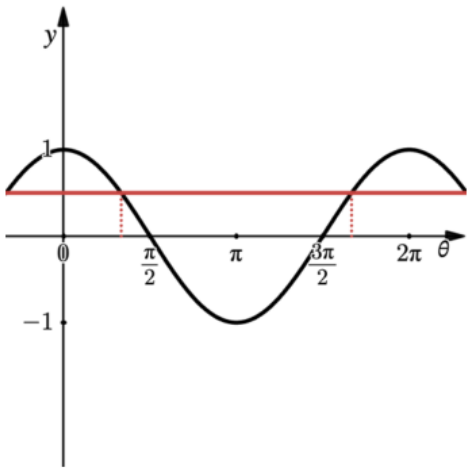
Find second value between 0 and 2π:

$\theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$ (in range).

Now solve $\cos\theta = \frac{2}{5}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$\theta = \arccos\frac{2}{5} = 1.1592\ldots$ (in range).



Find second value between 0 and 2π:

$\theta = 2\pi - 1.1592\ldots = 5.1239\ldots$ (in range).

$\Rightarrow x = \{1.16, \frac{7\pi}{6}, 5.12, \frac{11\pi}{6}\}$ to 3sf.

23 a)

$3\sin\theta\tan\theta = 2\sin\theta + \cos\theta$ for $0 \leq \theta < 2\pi$.

$$\Rightarrow 3\sin\theta \times \frac{\sin\theta}{\cos\theta} = 2\sin\theta + \cos\theta$$

$$\Rightarrow 3\sin^2\theta = 2\sin\theta\cos\theta + \cos^2\theta$$

$$\Rightarrow 3\sin^2\theta - 2\sin\theta\cos\theta - \cos^2\theta = 0$$

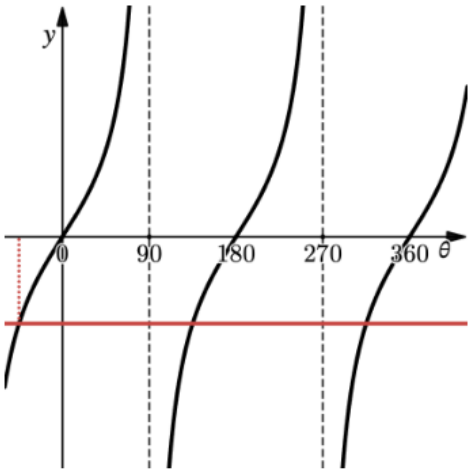
$$\Rightarrow (3\sin\theta + \cos\theta)(\sin\theta - \cos\theta) = 0.$$

First solve $3\sin\theta + \cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$$\Rightarrow \frac{3\sin\theta}{\cos\theta} + 1 = 0 \Rightarrow 3\tan\theta + 1 = 0 \Rightarrow \tan\theta = -\tfrac{1}{3}.$$

Find first value:

$\theta = \arctan(-\frac{1}{3}) = -0.3217\ldots$ (not in range).



Add on period:

$\theta = -0.3217\ldots + \pi = 2.8198\ldots$ (in range).

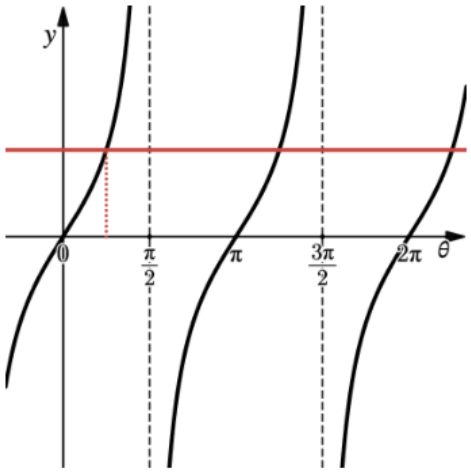
$\theta = 2.8198\ldots + \pi = 5.9614\ldots$ (in range).

Now solve $\sin\theta - \cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{\sin\theta}{\cos\theta} - 1 = 0 \Rightarrow \tan\theta - 1 = 0 \Rightarrow \tan\theta = 1.$

Find first value:

$\theta = \arctan 1 = \frac{\pi}{4}$ (in range).



Add period:

$\theta = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$ (in range).

$\Rightarrow \theta = \{\frac{\pi}{4}, 2.82, \frac{5\pi}{4}, 5.96\}$ to 3sf.

23 b)

$8\sin\theta\tan\theta + 39\sin\theta = 5\cos\theta$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 8\sin\theta \times \dfrac{\sin\theta}{\cos\theta} + 39\sin\theta = 5\cos\theta$

$\Rightarrow 8\sin^2\theta + 39\sin\theta\cos\theta = 5\cos^2\theta$

$\Rightarrow 8\sin^2\theta + 39\sin\theta\cos\theta - 5\cos^2\theta = 0$

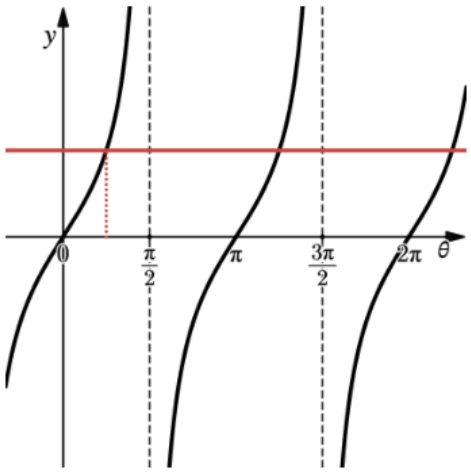
$\Rightarrow (8\sin\theta - \cos\theta)(\sin\theta + 5\cos\theta) = 0.$

First solve $8\sin\theta - \cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{8\sin\theta}{\cos\theta} - 1 = 0 \Rightarrow 8\tan\theta - 1 = 0 \Rightarrow \tan\theta = \frac{1}{8}.$

Find first value:

$\theta = \arctan\frac{1}{8} = 0.1243\ldots$ (in range).



Add period:

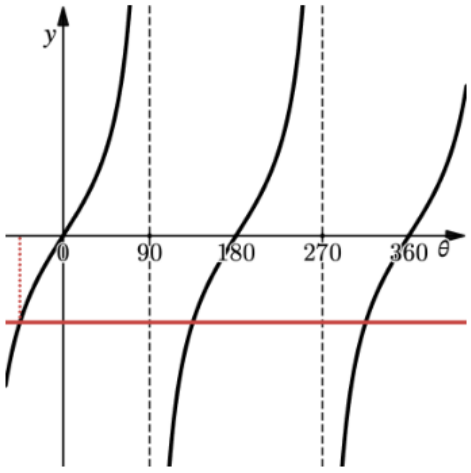
$\theta = 0.1243\ldots + \pi = 3.2659\ldots$ (in range).

Now solve $\sin\theta + 5\cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{\sin\theta}{\cos\theta} + 5 = 0 \Rightarrow \tan\theta + 5 = 0 \Rightarrow \tan\theta = -5.$

Find first value:

$\theta = \arctan(-5) = -1.3734\ldots$ (not in range).



Add on period:

$\theta = -1.3734\ldots + \pi = 1.7681\ldots$ (in range).

$\theta = 1.7681\ldots + \pi = 4.9097\ldots$ (in range).

$\Rightarrow \theta = \{0.124, 1.77, 3.27, 4.91\}$ to 3sf.

23 c)

$74\cos\theta\tan\theta = 45\sin\theta\tan\theta + 24\cos\theta$ for $0 \leq \theta < 2\pi$.

$\Rightarrow 74\cos\theta \times \dfrac{\sin\theta}{\cos\theta} = 45\sin\theta \times \dfrac{\sin\theta}{\cos\theta} + 24\cos\theta$

$\Rightarrow 74\sin\theta = \dfrac{45\sin^2\theta}{\cos\theta} + 24\cos\theta$

$\Rightarrow 74\sin\theta\cos\theta = 45\sin^2\theta + 24\cos^2\theta$

$\Rightarrow 45\sin^2\theta - 74\sin\theta\cos\theta + 24\cos^2\theta = 0$

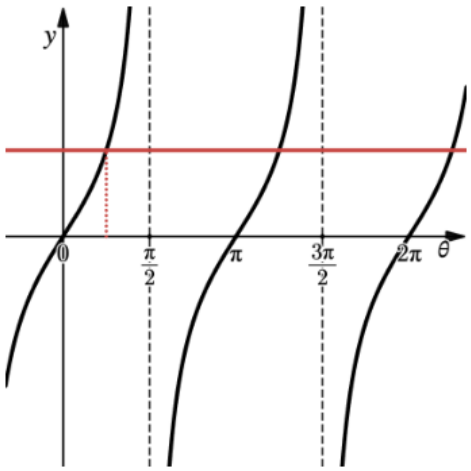
$\Rightarrow (9\sin\theta - 4\cos\theta)(5\sin\theta - 6\cos\theta) = 0.$

First solve $9\sin\theta - 4\cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{9\sin\theta}{\cos\theta} - 4 = 0 \Rightarrow 9\tan\theta - 4 = 0 \Rightarrow \tan\theta = \frac{4}{9}$.

Find first value:

$\theta = \arctan\frac{4}{9} = 0.4182\ldots$ (in range).



Add period:

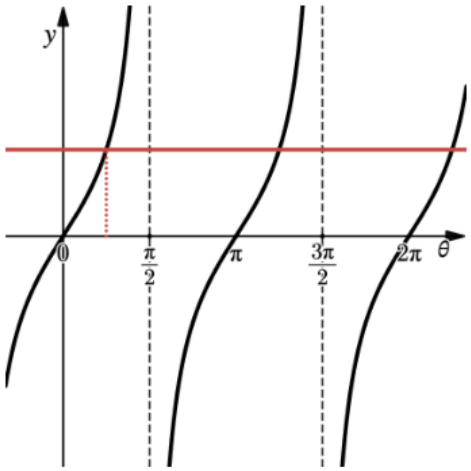
$\theta = 0.4182\ldots + \pi = 3.5598\ldots$ (in range).

Now solve $5\sin\theta - 6\cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{5\sin\theta}{\cos\theta} - 6 = 0 \Rightarrow 5\tan\theta - 6 = 0 \Rightarrow \tan\theta = \frac{6}{5}$.

Find first value:

$\theta = \arctan\frac{6}{5} = 0.8760\ldots$ (in range).



Add period:

$\theta = 0.8760\ldots + \pi = 4.0176\ldots$ (in range).

$\Rightarrow \theta = \{0.418, 0.876, 3.56, 4.02\}$ to 3sf.

23 d)

$40\sin\theta\tan\theta = 15\cos\theta + 38\cos\theta\tan\theta$ for $0 \leq \theta < 2\pi$.

$$\Rightarrow 40\sin\theta \times \frac{\sin\theta}{\cos\theta} = 15\cos\theta + 38\cos\theta \times \frac{\sin\theta}{\cos\theta}$$

$$\Rightarrow \frac{40\sin^2\theta}{\cos\theta} = 15\cos\theta + 38\sin\theta$$

$$\Rightarrow 40\sin^2\theta = 15\cos^2\theta + 38\sin\theta\cos\theta$$

$$\Rightarrow 40\sin^2\theta - 38\sin\theta\cos\theta - 15\cos^2\theta = 0$$

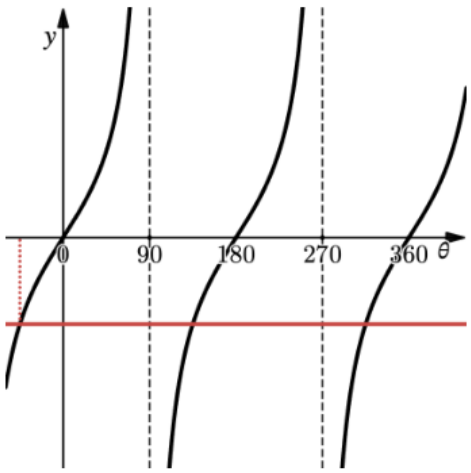
$$\Rightarrow (10\sin\theta + 3\cos\theta)(4\sin\theta - 5\cos\theta) = 0.$$

First solve $10\sin\theta + 3\cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$$\Rightarrow \frac{10\sin\theta}{\cos\theta} + 3 = 0 \Rightarrow 10\tan\theta + 3 = 0 \Rightarrow \tan\theta = -\tfrac{3}{10}.$$

Find first value:

$\theta = \arctan(-\frac{3}{10}) = -0.2914\ldots$ (not in range).



Add on period:

$\theta = -0.2914\ldots + \pi = 2.8501\ldots$ (in range).

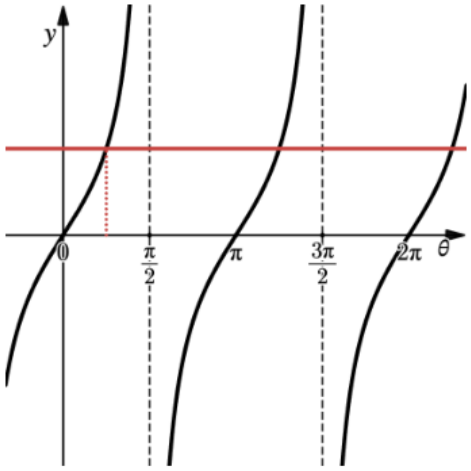
$\theta = 2.8501\ldots + \pi = 5.9917\ldots$ (in range).

Now solve $4\sin\theta - 5\cos\theta = 0$ for $0 \leq \theta < 2\pi$.

$\Rightarrow \dfrac{4\sin\theta}{\cos\theta} - 5 = 0 \Rightarrow 4\tan\theta - 5 = 0 \Rightarrow \tan\theta = \frac{5}{4}$.

Find first value:

$\theta = \arctan\frac{5}{4} = 0.8960\ldots$ (in range).



Add period:

$\theta = 0.8960\ldots + \pi = 4.0376\ldots$ (in range).

$\Rightarrow \theta = \{0.896, 2.85, 4.04, 5.99\}$.

24 a)

$18\sin^2 x + 3\cos x = 8$ for $0 \leq x < 4\pi$.

$\Rightarrow 18(1 - \cos^2 x) + 3\cos x = 8$

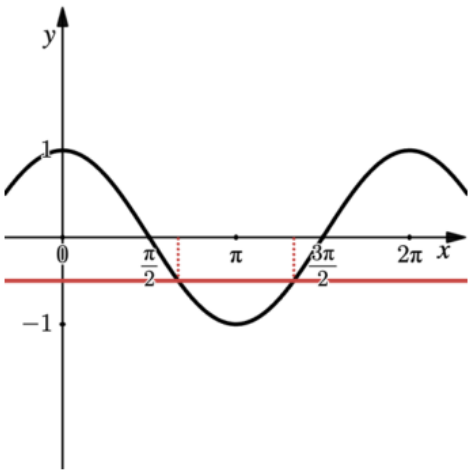
$\Rightarrow 18 - 18\cos^2 x + 3\cos x = 8$

$\Rightarrow 18\cos^2 x - 3\cos x - 10 = 0 \Rightarrow \cos x = -\frac{2}{3}$ or $\cos x = \frac{5}{6}$.

First solve $\cos x = -\frac{2}{3}$ for $0 \leq x < 4\pi$.

Find first value between 0 and 2π:

$x = \arccos(-\frac{2}{3}) = 2.3005\ldots$ (in range).



Find second value between 0 and 2π:

$x = 2\pi - 2.3005\ldots = 3.9826\ldots$ (in range).

Add period:

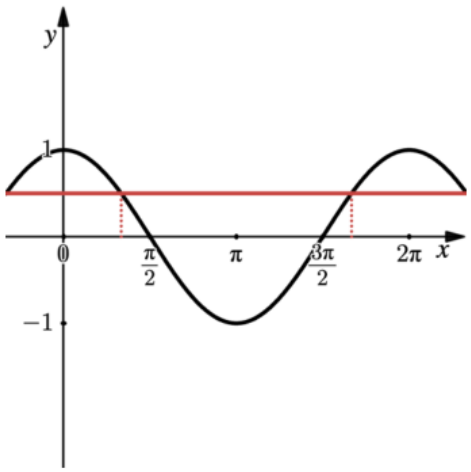
$x = 2.3005\ldots + 2\pi = 8.5837\ldots$ (in range).

$x = 3.9826\ldots + 2\pi = 10.2658\ldots$ (in range).

Now solve $\cos x = \frac{5}{6}$ for $0 \leq x < 2\pi$.

Find first value between 0 and 2π:

$x = \arccos\frac{5}{6} = 0.5856\ldots$ (in range).



Find second value between 0 and 2π:

$x = 2\pi - 0.5856\ldots = 5.6974\ldots$ (in range).

Add period:

$x = 0.5856\ldots + 2\pi = 6.8688\ldots$ (in range).

$x = 5.6974\ldots + 2\pi = 11.9806\ldots$ (in range).

$\Rightarrow x = \{0.586, 2.30, 3.98, 5.70, 6.87, 8.58, 10.3, 12.0\}$ to 3sf.

24 b)

$\frac{1}{3}\theta = 0.5856\ldots \Rightarrow \theta = 1.7570\ldots$

$\frac{1}{3}\theta = 2.3005\ldots \Rightarrow \theta = 6.9015\ldots$

$\frac{1}{3}\theta = 3.9826\ldots \Rightarrow \theta = 11.9479\ldots$

$\frac{1}{3}\theta = 5.6974\ldots \Rightarrow \theta = 17.0924\ldots$

$\frac{1}{3}\theta = 6.8688\ldots \Rightarrow \theta = 20.6066\ldots$

$\frac{1}{3}\theta = 8.5837\ldots \Rightarrow \theta = 25.7511\ldots$

$\frac{1}{3}\theta = 10.2658\ldots \Rightarrow \theta = 30.7975\ldots$

$\frac{1}{3}\theta = 11.9806\ldots \Rightarrow \theta = 35.9420\ldots$

$\Rightarrow \theta = \{1.76, 6.90, 11.9, 17.1, 20.6, 25.8, 30.8, 35.9\}$ to 3sf.

25 a)

LHS:

$$\begin{aligned}\frac{2\sin x - 5\cos^2 x - 2}{5\sin x + 7} &\equiv \frac{2\sin x - 5(1 - \sin^2 x) - 2}{5\sin x + 7} \\ &\equiv \frac{2\sin x - 5 + 5\sin^2 x - 2}{5\sin x + 7} \equiv \frac{5\sin^2 x + 2\sin x - 7}{5\sin x + 7} \\ &\equiv \frac{(5\sin x + 7)(\sin x - 1)}{5\sin x + 7} \equiv \sin x - 1 \quad \text{:RHS.}\end{aligned}$$

25 b)

$\frac{2\sin(\theta+\frac{\pi}{3})-5\cos^2(\theta+\frac{\pi}{3})-2}{5\sin(\theta+\frac{\pi}{3})+7}=10\cos(\theta+\frac{\pi}{3})-1$ for $0\leq\theta<2\pi$.

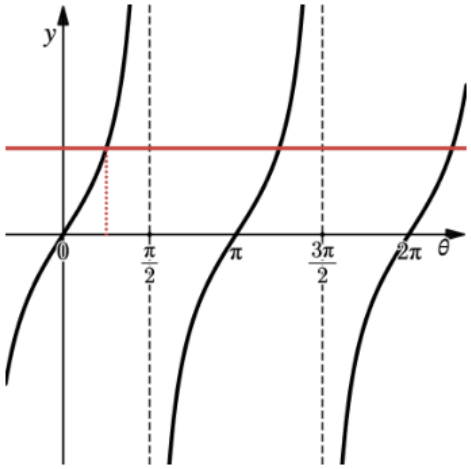
$\Rightarrow \sin(\theta+\frac{\pi}{3})-1=10\cos(\theta+\frac{\pi}{3})-1$

$\Rightarrow \sin(\theta+\frac{\pi}{3})=10\cos(\theta+\frac{\pi}{3})\Rightarrow\frac{\sin(\theta+\frac{\pi}{3})}{\cos(\theta+\frac{\pi}{3})}=10$

$\Rightarrow \tan(\theta+\frac{\pi}{3})=10.$

Find first value:

$\theta+\frac{\pi}{3}=\arctan 10=1.4711\ldots$



Apply transformation:

$\theta+\frac{\pi}{3}=1.4711\ldots\Rightarrow\theta=0.4239\ldots$ (in range).

Add period:

$\theta=0.4239\ldots+\pi=3.5655\ldots$ (in range).

$\theta=\{0.424,3.57\}$ to 3sf.

26 a)

LHS:

$$\begin{aligned}\frac{25-40\cos^2 x-38\sin x}{4\sin x-5}&\equiv\frac{25-40(1-\sin^2 x)-38\sin x}{4\sin x-5}\\&\equiv\frac{25-40+40\sin^2 x-38\sin x}{4\sin x-5}\equiv\frac{40\sin^2 x-38\sin x-15}{4\sin x-5}\\&\equiv\frac{(4\sin x-5)(10\sin x+3)}{4\sin x-5}\equiv 3+10\sin x\quad\text{:RHS.}\end{aligned}$$

26 b)

$\dfrac{25 - 40\cos^2 2\theta - 38\sin 2\theta}{4\sin 2\theta - 5} = 3 - 8\cos 2\theta$ for $0° \leq \theta < 180°$.

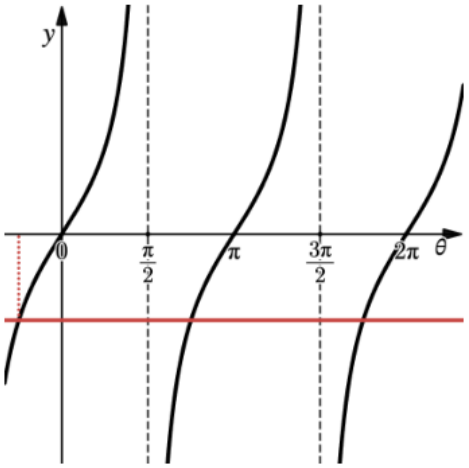
$\Rightarrow 3 + 10\sin 2\theta = 3 - 7\cos 2\theta$

$\Rightarrow 10\sin 2\theta = -8\cos 2\theta \Rightarrow \dfrac{10\sin 2\theta}{\cos 2\theta} = -8$

$\Rightarrow 10\tan 2\theta = -8 \Rightarrow \tan 2\theta = -\frac{4}{5}$.

Find first value:

$2\theta = \arctan(-\frac{4}{5}) = -0.6747\ldots$



Apply transformation:

$2\theta = -0.6747\ldots \Rightarrow \theta = -0.3373\ldots$ (not in range).

Add period:

$\theta = -0.3373\ldots + \frac{\pi}{2} = 1.2334\ldots$ (in range).

$\theta = 1.2334\ldots + \frac{\pi}{2} = 2.8042\ldots$ (in range).

$\theta = \{1.23, 2.80\}$ to 3sf.

27)

$\sqrt{18\sin^2\theta - 21\sin\theta\cos\theta} = 2\cos\theta$ for $0 \le \theta < 2\pi$.

$\Rightarrow 18\sin^2\theta - 21\sin\theta\cos\theta = 4\cos^2\theta$

$\Rightarrow 18\sin^2\theta - 21\sin\theta\cos\theta - 4\cos^2\theta = 0$

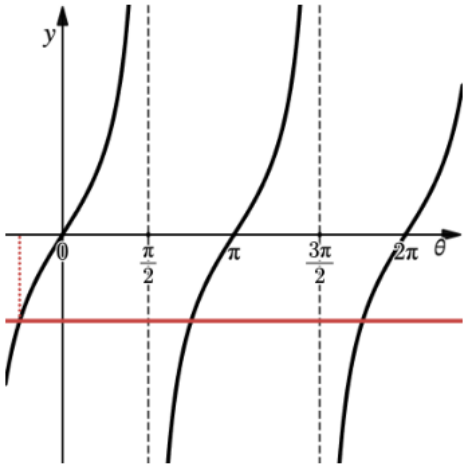
$\Rightarrow (6\sin\theta + \cos\theta)(3\sin\theta - 4\cos\theta) = 0.$

First solve $6\sin\theta + \cos\theta = 0$ for $0 \le \theta < 2\pi$.

$\Rightarrow \dfrac{6\sin\theta}{\cos\theta} + 1 = 0 \Rightarrow 6\tan\theta + 1 = 0 \Rightarrow \tan\theta = -\frac{1}{6}.$

Find first value:

$\theta = \arctan(-\frac{1}{6}) = -0.1651\ldots$



Add period:

$\theta = -0.1651\ldots + \pi = 2.9764\ldots$ (in range).

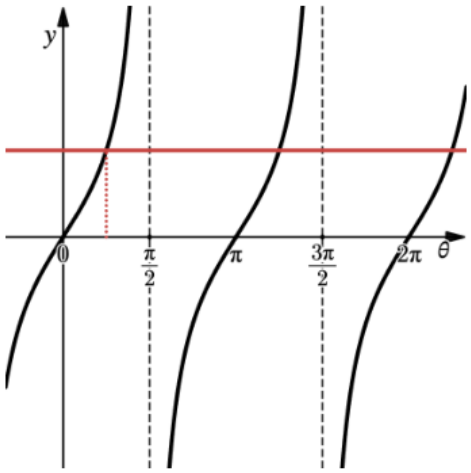
$\theta = 2.9764\ldots + \pi = 6.1180\ldots$ (in range).

Now solve $3\sin\theta - 4\cos\theta = 0$ for $0 \le \theta < 2\pi$.

$\Rightarrow \dfrac{3\sin\theta}{\cos\theta} - 4 = 0 \Rightarrow 3\tan\theta - 4 = 0 \Rightarrow \tan\theta = \frac{4}{3}.$

Find first value:

$\theta = \arctan\frac{4}{3} = 0.9272\ldots$



Add period:

$\theta = 0.9272\ldots + \pi = 4.0688\ldots$ (in range).

$\theta = \{0.927, 2.98, 4.07, 6.12\}$ to 3sf.

The original equation was $\sqrt{18\sin^2\theta - 21\sin\theta\cos\theta} = 2\cos\theta$.

The left-hand side is ≥ 0 so only solutions where $2\cos\theta \geq 0$ solve the equation.

$2\cos\theta \geq 0$ occurs when $0 \leq \theta \leq \frac{\pi}{2}$ or when $\frac{3\pi}{2} \leq \theta \leq 2\pi$, so 2.98 and 4.07 do not solve the original equation.

$\therefore \theta = \{0.927, 6.12\}$ to 3sf.

28 a)

RHS:

$$\begin{aligned}\frac{\sin\theta\cos\theta}{\frac{1}{\tan\theta} - \tan\theta} &\equiv \frac{\sin\theta\cos\theta}{\frac{1}{\frac{\sin\theta}{\cos\theta}} - \frac{\sin\theta}{\cos\theta}} \equiv \frac{\sin\theta\cos\theta}{\frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta}} \\ &\equiv \frac{\sin^2\theta\cos^2\theta}{\cos^2\theta - \sin^2\theta} \equiv \frac{(1-\cos^2\theta)\cos^2\theta}{\cos^2\theta - (1-\cos^2\theta)} \\ &\equiv \frac{\cos^2\theta - \cos^4\theta}{\cos^2\theta - 1 + \cos^2\theta} \equiv \frac{\cos^2\theta - \cos^4\theta}{2\cos^2\theta - 1} \quad \text{:LHS.}\end{aligned}$$

28 b)

RHS:

$$\begin{aligned}\frac{\sin\theta\cos\theta}{\frac{1}{\tan\theta} - \tan\theta + \cos\theta} &\equiv \frac{\sin\theta\cos\theta}{\frac{1}{\frac{\sin\theta}{\cos\theta}} - \frac{\sin\theta}{\cos\theta} + \cos\theta} \\ &\equiv \frac{\sin\theta\cos\theta}{\frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} + \cos\theta} \equiv \frac{\sin^2\theta\cos^2\theta}{\cos^2\theta - \sin^2\theta + \sin\theta\cos^2\theta} \\ &\equiv \frac{\sin^2\theta(1-\sin^2\theta)}{(1-\sin^2\theta) - \sin^2\theta + \sin\theta(1-\sin^2\theta)} \\ &\equiv \frac{\sin^2\theta - \sin^4\theta}{1 - \sin^2\theta - \sin^2\theta + \sin\theta - \sin^3\theta} \equiv \frac{\sin^2\theta - \sin^4\theta}{1 + \sin\theta - 2\sin^2\theta - \sin^3\theta} \\ &\equiv \frac{\sin^4\theta - \sin^2\theta}{\sin^3\theta + 2\sin^2\theta - \sin\theta - 1} \quad \text{:LHS.}\end{aligned}$$